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Aria-NBV: Quality-Driven Next-Best-View Planning with Egocentric Foundation Models

TODO: Deutscher Titel

Master's Thesis / Masterarbeit

for obtaining the academic degree Master of Science (M.Sc.)

submitted to
Munich University of Applied Sciences
Department of Computer Science & Mathematics

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Program: Informatik - Master
Specialization: Visual Computing and Machine Learning
Matriculation Number: 00807819

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Submission Date: TODO: Abgabedatum

Transparency in the Use of AI Tools

AI-assisted tools were used to organize literature notes, check consistency across repository documentation, and draft parts of the thesis seed. The author remains responsible for the final research scope, technical claims, citations, implementation, experiments, and submitted document.

Validation TODO: Update this statement against final institutional requirements before submission.

Source: proposal transparency text

Acknowledgements

Open question: Fill acknowledgements close to submission.

Source: main thesis seed

Abstract

This thesis investigates target-conditioned, quality-driven Next-Best-View (NBV) planning for egocentric 3D reconstruction in Aria Synthetic Environments (ASE). It uses target-specific Relative Reconstruction Improvement (RRI) as the supervision and evaluation signal for finite candidate view selection, measures bounded oracle-lookahead headroom, and tests whether an actor-visible finite-horizon value model can recover part of that headroom under matched oracle re-evaluation.

Zusammenfassung

Open question: Write the German abstract after the English thesis claim and final evidence scale are stable.

Source: main thesis seed

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Glossary and Abbreviations

Dataset

ADT – Aria Digital Twin: Project Aria dataset with real-world captures and digital-twin scene annotations. ADT is adjacent to ARIA-NBV's sim-to-real context; the current experiments focus on ASE and EFM3D/ATEK exports, while ADT remains a relevant transfer surface.

AEO – Aria Everyday Objects: Small-scale real-world Project Aria object dataset used by EFM3D for egocentric 3D perception evaluation. AEO is relevant as a possible sim-to-real check for ARIA-NBV ideas that are first developed on ASE mesh-supervised snippets.

ASE – Aria Synthetic Environments: Large-scale synthetic indoor dataset with simulated Project Aria sensor characteristics, egocentric trajectories, and scene annotations. ARIA-NBV uses ASE snippets and the public mesh-supervised subset to generate oracle RRI labels from ground-truth meshes and semi-dense point clouds.

CPF – Central Pupil Frame: Coordinate frame placed at the midpoint between the left and right eye boxes of Project Aria glasses. CPF is used by Project Aria for gaze-related quantities and should remain distinct from rig, camera, world, and PyTorch3D frames in ARIA-NBV docs.

MPS – Machine Perception Services: Project Aria processing services that derive pose, mapping, gaze, hand, and related perception outputs from sensor recordings. ARIA-NBV mainly depends on MPS-style pose and semi-dense mapping products as the current reconstruction state for RRI computation.

MTD – Motion Trajectory Data: Device poses over time, usually represented as a sequence of 6-DoF transformations. MTD supplies the logged egocentric trajectory used to define snippet state, current reconstruction context, and candidate-view reference poses.

MFCD – Multi-Frame Camera Data: Synchronized camera streams from multiple Project Aria cameras over a temporal window. MFCD provides the

egocentric image evidence consumed by EVL and aligned with pose, calibration, and semi-dense point streams.

MSDPD – Multi-Semi-Dense Point Data: Semi-dense 3D point observations generated by SLAM-style processing across a snippet or trajectory window. MSDPD is the sparse observed geometry that ARIA-NBV fuses with candidate point clouds and compares against ground-truth meshes for RRI labels.

Project Aria – Project Aria: Meta's egocentric research-device and tooling ecosystem for calibrated, time-aligned multimodal sensing. ARIA-NBV treats Project Aria and its MPS-style products as the actor-visible sensing contract, while ASE meshes remain offline supervision and evaluation assets.

SSL – SceneScript Language: Structured language representation for indoor scene layout using primitives such as walls, doors, windows, and objects. SceneScript is relevant to ARIA-NBV as a possible semantic/global planning layer and as one source of ASE scene-structure context.

snippet – Snippet: Short synchronized temporal window of Aria sensor data used as one EVL/VIN input sample. A snippet typically contains RGB or grayscale streams, poses, calibration, semi-dense points, and scene metadata that EVL lifts into a voxel grid.

VRS – Virtual Reality Standard: File format used by Project Aria tooling to store multi-modal sensor recordings efficiently. VRS is part of the upstream Project Aria ecosystem, while ARIA-NBV primarily consumes ASE/ATEK-derived tensorized snippets for current experiments.

Entity

target – Target of Interest: Selected entity, object crop, point, region, or surface-deficit hypothesis whose reconstruction quality should be improved. Target-conditioned ARIA-NBV variants use this target as an explicit input to candidate scoring and planning instead of optimizing only scene-level RRI.

Geometry

***DoF* – Degrees of Freedom:** Number of independent pose parameters available to a camera, object, or action representation. ARIA-NBV distinguishes full 6-DoF pose estimation from reduced 5-DoF candidate or action spaces used for practical view planning.

***frustum* – Frustum:** Truncated pyramidal camera-visible volume bounded by near and far clipping planes plus lateral field-of-view planes. Candidate-view frusta define which scene surfaces can project into a camera image and are therefore central to visibility, rendering, and RRI diagnostics.

***LUF* – Left-Up-Forward:** Camera coordinate convention whose x axis points left, y axis points up, and z axis points forward. LUF is one of the coordinate-frame conventions that must stay explicit when moving between Project Aria cameras, PyTorch3D cameras, and ARIA-NBV candidate poses.

***OBB* – Oriented Bounding Box:** 3D bounding box with arbitrary orientation, used to represent object extent more tightly than an axis-aligned box. OBBs are a natural target encoding for entity-aware ARIA-NBV because they provide center, extent, orientation, semantic class, and confidence signals.

***6DoF* – Six Degrees of Freedom:** Pose parameterization with three translational and three rotational degrees of freedom. Project Aria poses, candidate cameras, and world-to-device transforms are usually represented as 6-DoF rigid-body transforms.

Metrics

***AUC* – Area Under Curve:** Aggregate score computed by integrating a metric curve over an acquisition, threshold, or ranking axis. AUC can summarize how quickly reconstruction quality, coverage, or ranking quality improves as views are acquired or candidate thresholds change.

***CD* – Chamfer Distance:** Historical bidirectional distance family used to compare reconstructed points against reference geometry. Thesis-facing ARIA-

NBV notation uses point-mesh error D with directional components $D_{P \rightarrow M}$ and $D_{M \rightarrow P}$; older seminar material may still call this CD.

CR – Coverage Ratio: Fraction of a target surface, scene, or region treated as observed under a chosen visibility or distance threshold. Coverage ratio is a useful diagnostic baseline for NBV, but ARIA-NBV treats reconstruction-quality improvement through RRI as the preferred optimization target.

GT – Ground Truth: Reference data or annotations treated as the trusted target for training, validation, or evaluation. In ARIA-NBV, ground truth usually refers to ASE meshes, object annotations, poses, or labels used to compute oracle supervision and diagnostic metrics.

RRI – Relative Reconstruction Improvement: Metric quantifying the relative reconstruction-quality improvement obtained by adding a candidate observation to the current reconstruction. ARIA-NBV computes RRI by comparing reconstruction error before and after fusing candidate-view geometry, usually against an ASE ground-truth mesh for oracle supervision and evaluation.

reward – Target-RRI Reward: Quality-only immediate reward for thesis-core target-aware rollouts. It is the target-specific point-mesh error reduction normalized by the rollout-root target error; scalar motion, rule, diversity, and validity penalties are later ablations.

target RRI – Target-Specific RRI: RRI computed only on the ground-truth and reconstructed geometry associated with a selected target of interest. Target-specific RRI lets the thesis compare views by how much they improve a selected object or region, even when scene-level RRI would prefer large background surfaces.

Model

EFM3D – Egocentric Foundation Model 3D: Egocentric 3D foundation-model stack used as the frozen spatial backbone for ARIA-NBV candidate scoring. The current project uses EFM3D and its EVL architecture to expose

voxel occupancy, centerness, semantic, and OBB evidence for VIN-style RRI prediction.

EVL – Egocentric Voxel Lifting: EFM3D architecture that lifts synchronized egocentric observations into a gravity-aligned 3D voxel feature volume. ARIA-NBV uses EVL head outputs, pre-head features, and OBB predictions as local actor-visible evidence and target support, not as the complete frozen scene representation.

Q_H – Finite-Horizon Q Function: Target-conditioned finite-candidate value function trained from ASE oracle rollout traces. The first thesis implementation uses candidate-to-state query attention and emits one masked continuous-return value per valid candidate-table action, using DQN-style replayed Bellman targets, Double-DQN selector/evaluator backups, and offline support constraints.

target-conditioned scorer – Target-Conditioned Scorer: VIN-style candidate scorer that receives scene state, a candidate view, and an encoding of the target of interest. The scorer predicts target-specific utility so view ranking can prioritize a selected entity or region instead of only optimizing scene-level RRI.

VIN – View Introspection Network: Learned candidate-view scorer that predicts RRI or an ordinal RRI-derived utility without capturing the candidate view. ARIA-NBV adapts VIN-NBV by placing a lightweight RRI prediction head on top of frozen EVL features and candidate-pose evidence from ASE snippets.

Planning

cost – Acquisition Cost: Budget consumed to acquire observations, measured by view count, path length, elapsed time, invalid-action rate, or a weighted combination. The thesis should first report cumulative root-normalized target gain, diagnostic target RRI, and acquisition cost separately, then use scalarized objectives only when the tradeoff is explicit.

candidate – Candidate View: Proposed camera pose whose expected reconstruction utility is evaluated before selecting the next observation. ARIA-NBV samples candidate views around a reference pose or target, renders candidate depths from the mesh for oracle labels, and scores candidates with RRI or learned VIN predictions.

transition – Counterfactual Transition: Deterministic or replayable rollout update that renders or retrieves only the selected candidate observation, backprojects it to candidate geometry, and fuses it into the accumulated reconstruction proxy.

action set – Finite Candidate Action Set: Valid discrete action set for ARIA-NBV planning, obtained by masking a sampled candidate view table. The action is the candidate-table index, not a continuous pose; the selected pose is $q_t = q_{t,a_t}$. Invalid candidates are constraints, not low-quality actions.

return – Finite-Horizon Return: Symbolic H-step return used by ARIA-NBV rollout evaluation and Q_H targets. The formula keeps γ explicit while the thesis-core comparison reports cumulative root-normalized target gain under equal acquisition budget.

5DoF – Five Degrees of Freedom: Reduced camera-action parameterization commonly used when roll is fixed or otherwise constrained. ARIA-NBV uses 5-DoF language for candidate and planning abstractions where position and viewing direction matter while roll is not an independently optimized control dimension.

CF+ state – Geometry-Rich Counterfactual State: Ablation actor state that extends the minimal counterfactual state with selected prior synthetic observations only: rendered depth, valid mask, backprojected point cloud, and derived local normals or support summaries.

historic state – Historic Snippet State: Raw actor-visible state available along the logged Project Aria/ASE trajectory before counterfactual rollout synthesis. It includes captured camera streams, calibration, timestamps, trajectory/gravity, semidense points with support or uncertainty, EVL/EFM

evidence, and detected or predicted OBBs; GT mesh and GT OBBs are excluded as actor inputs.

CFO state – Minimal Counterfactual Actor State: Main thesis-core actor-visible counterfactual rollout state. It keeps the fused point proxy $\mathcal{P}_t$ as broad scene state, optional lifted image-foundation features attached to points, local root EVL evidence for target support, selected-view history, target descriptor, budget, candidate table, validity masks and reason codes, and current candidate-query features, without exposing all-candidate GT renders.

NBV – Next-Best-View: Problem of selecting the next sensor viewpoint to improve an active reconstruction or inspection objective under a limited acquisition budget. In ARIA-NBV, the preferred objective is reconstruction quality improvement rather than coverage alone, with candidate views ranked by oracle or learned RRI scores.

oracle state – Oracle Rollout State: Privileged state used only for ASE mesh-supervised label generation, upper-bound planning, and evaluation. It includes GT mesh and target crops, all-candidate renders, points, normals, mesh-face visibility, RRI and point-mesh error metrics, oracle scores, and GT OBBs.

offline state – Persisted Offline Sample State: Compact VIN offline-store state persisted for training and diagnostics. It contains the VinSnippetView payload, candidate table and cameras, candidate count, validity or count metadata, oracle metrics as labels, optional rendered depths, compact OBBs, trajectory metadata, and selected EVL numeric fields.

state – Rollout State: Family of rollout state variants used by ARIA-NBV. The actor-visible variants contain logged or replayed observations, reconstruction proxies, candidates, validity masks, target descriptors, history, and budget; the oracle variant adds privileged GT geometry and all-candidate labels for supervision and evaluation only.

NBV MDP – Target-Conditioned NBV MDP: Finite-horizon Markov decision process used to define ARIA-NBV’s target-conditioned rollout and fitted Q_H training contract. The actor observes only current reconstruction state, sampled candidate views, validity masks, target descriptors, history, and budget state; oracle meshes and GT crops are supervision and evaluation signals.

mask – Validity Mask: Hard candidate-level feasibility mask used during candidate ranking, rollout generation, and Q_H training. Invalidity is represented separately from reconstruction quality and should not be encoded as the lowest RRI class.

Protocol

GT-EVAL – Ground-Truth Target Evaluation: Main thesis protocol component restricting ground-truth OBBs and target mesh crops to oracle labels, target-RRI computation, matching checks, and evaluation. GT target crops are not actor-visible inputs in the main result.

OBS-SEL – Observed Target Selection: Main thesis protocol component requiring target selection to use only actor-visible evidence such as predicted or tracked OBBs, class probabilities, confidence, projected area, and semidense or EVL support. Ground-truth target annotations are not visible to the selector in this protocol.

PRED-Q – Predicted-Target Q: Main thesis protocol component requiring the scorer or fitted Q_H model to condition on predicted or observed target descriptors rather than ground-truth target inputs. It covers target-conditioned one-step scoring and finite-candidate Q_H selection.

Reconstruction

MVS – Multi-view Stereo: Reconstruction family that estimates dense or semi-dense scene geometry from multiple calibrated views. MVS is part of the broader reconstruction context for NBV, although ARIA-NBV’s current

oracle labels are built from ASE meshes and Aria-style semi-dense point observations.

PC – Point Cloud: Set of 3D points representing observed scene geometry. ARIA-NBV compares current and candidate-fused point clouds against ground-truth meshes to compute oracle RRI and surface-distance diagnostics.

SLAM – Simultaneous Localization and Mapping: Method for estimating sensor motion while building a map of the surrounding scene. In ARIA-NBV, logged SLAM poses and semi-dense points form the current reconstruction state against which candidate-view RRI is evaluated.

track – Track: Temporal sequence of corresponding image-feature detections across frames, usually carrying per-frame image coordinates, timestamps, and camera IDs. Tracks can be triangulated or optimized into 3D points, making them a bridge between image evidence and the semi-dense point clouds used by ARIA-NBV.

VIO – Visual-Inertial Odometry: Pose-estimation method that combines visual measurements from cameras with inertial measurements from IMUs. Project Aria pose and mapping products build on visual-inertial estimation, and ARIA-NBV depends on those pose streams for snippet state and candidate frame contracts.

Representation

Occupancy Grid – Occupancy Grid: Spatial grid whose cells encode whether space is occupied, free, unknown, or represented by a related occupancy probability. Occupancy-style voxel evidence appears in EVL outputs and VIN feature construction, where it helps summarize local 3D scene state for candidate scoring.

3DGS – 3D Gaussian Splatting: Explicit radiance-field representation using optimized 3D Gaussian primitives for real-time novel-view synthesis. ARIA-NBV treats active 3DGS work as proposal, uncertainty, and simulator-bridge literature, not as a replacement for ASE mesh-supervised RRI.

Supervision

oracle RRI – **Oracle RRI**: RRI label computed with privileged ground-truth geometry, used for supervised training and evaluation. The current oracle renders candidate depth maps from ASE ground-truth meshes, backprojects candidate point clouds, fuses them with the current semi-dense reconstruction, and scores the resulting surface-distance improvement.

List of Symbols

Symbol	Meaning	Key
Oracle and Reconstruction		
RRI	Relative Reconstruction Improvement score for a candidate view.	oracle.rri
$\mathcal{P}$	Actor-visible point set accumulated from observed or replayed views.	oracle.points
$\mathcal{P}_q$	Candidate-view point contribution used in oracle or counterfactual updates.	oracle.points_q
$\mathcal{Q}$	Finite candidate-view set considered by the planner.	oracle.candidates
$\mathcal{Q}_t$	Candidate-view set available at rollout step t.	oracle.candidates_t
$q_{t,i}$	Candidate pose i at rollout step t.	oracle.candidate_qti
D_q	Oracle-rendered or candidate-specific depth map for a proposed view.	oracle.depth_q
$D_{P \rightarrow M}$	Point-to-mesh directional error component.	oracle.acc
$D_{M \rightarrow P}$	Mesh-to-point directional error component.	oracle.comp
RRI Metrics		
D	Aggregate point-mesh reconstruction error used by RRI definitions.	rri.cd_value
ASE Assets		
$\mathcal{M}^{\text{GT}}$	Ground-truth scene mesh used for oracle labels and evaluation.	ase.mesh
$\mathcal{M}_e^{\text{GT}}$	Ground-truth mesh crop for target e.	ase.mesh_target
$\mathcal{F}^{\text{GT}}$	Ground-truth mesh faces used in point-mesh distance calculations.	ase.faces
$T_{\text{rig}}^{\text{w}}(t)$	Project Aria rig pose in the world frame at time t.	ase.traj
$T_{\text{rig}}^{\text{w}}(T)$	Final rig pose used as an anchor for gravity-aligned local fields.	ase.traj_final
VIN Scorer		
r	Target or scene RRI value used as a model target.	vin.rri
$\hat{r}$	Predicted RRI value emitted by the learned scorer.	vin.rri_hat
$\mathcal{L}$	Training loss for scorer or value-model objectives.	vin.loss
m	Candidate validity indicator used by scorer diagnostics.	vin.cand_valid
Entity and Target		
RRI_e	Target-specific Relative Reconstruction Improvement for entity e.	entity.rri_e

Symbol	Meaning	Key
φ_e	Actor-visible target/entity descriptor used to condition target-specific value prediction.	entity.target_desc
Planning and Rollout		
s	Abstract planning state.	rl.s
a	Action index selecting a candidate row.	rl.a
r	Scalar reward or immediate gain.	rl.r
G	Finite-horizon return accumulated across rollout steps.	rl.G
$\mathcal{M}_{\text{NBV}}$	Target-conditioned finite-candidate NBV decision process.	rl.mdp_nbv
$\mathcal{A}(s_t)$	Valid action set available in state s_t .	rl.action_set
T	State-transition operator for selected candidate actions.	rl.transition
s_t^{hist}	Logged historic snippet state at rollout step t .	rl.s_hist
s_t^{off}	Persisted offline sample state used by training readers.	rl.s_off
$s_t^{\text{cf}0}$	Minimal actor-visible counterfactual state.	rl.s_cf0
$s_t^{\text{cf}+}$	Geometry-enriched counterfactual successor state.	rl.s_cf_geom
s_t^{oracle}	Oracle-only state used for labels, upper bounds, and evaluation.	rl.s_oracle
r_t^e	Target-specific reward at rollout step t .	rl.reward_target
$G_t^{(H)}$	H-step finite-horizon return from rollout step t .	rl.return_h
Q_H	Finite-horizon candidate-value function.	rl.qh
γ	Return discount factor.	rl.gamma
H	Planning or rollout horizon length.	rl.H
$m_{t,i}$	Hard validity mask for candidate i at rollout step t .	rl.validity_mask
$\rho_{t,i}$	Invalid-action reason code for candidate i at rollout step t .	rl.invalid_reason
e_t	Selected target entity at rollout step t .	rl.target
b_t	Remaining acquisition budget at rollout step t .	rl.budget
$C(\tau)$	Acquisition cost of a selected trajectory.	rl.acquisition_cost
Shape and Size		
N_q	Number of candidate views in a candidate table.	shape.Nq
K	Number of ordinal bins or discrete levels when used in scorer losses.	shape.K
scene		
M_t^{ray}	Sparse actor-visible ray-aware memory separating occupied, free, and unknown support at step t .	scene.ray_memory_t
Observation and Actor State		

Symbol	Meaning	Key
$F_t^{\text{DINO@pt}}$	Visibility-gated logged DINO feature bank attached to semidense or fused world points.	obs.dino_point_bank_t
X_t^{pt}	Point-token set combining geometry, support, uncertainty, history, and optional compressed logged descriptors.	obs.point_tokens_t
scene		
Φ_t^{scene}	Composite actor-visible scene-memory descriptor queried by the finite-candidate value model.	scene.scene_memory_t
$E_0^{\text{EVL-local}}$	Root local EVL evidence field used for target/candidate support reads.	scene.evl_local
$\omega_{t,i}^{\text{EVL}}$	Candidate or target-query fraction supported by the root EVL voxel extent.	scene.evl_support_frac
$g_{t,i}^{\text{EVL}}$	Pooled local EVL evidence token for a target, candidate frustum, or target-candidate intersection query.	scene.evl_support_token
g_e^{tgt}	Pooled target-support descriptor for the selected target hypothesis.	scene.target_support_pool
$g_{t,i}^{\text{fr}}$	Candidate-frustum support descriptor pooled from actor-visible scene memory.	scene.frustum_support_pool
$g_{t,e,i}^{\text{cap}}$	Pooled descriptor over the intersection between target support and candidate frustum support.	scene.target_frustum_pool
$g_{t,i}^{\text{ray}}$	Candidate-conditioned ray query over occupied, free, unknown, hit, target-support, and uncertainty summaries.	scene.ray_query_ti
RenderQuery	Abstract query operator that summarizes scene memory in a candidate camera/frustum without introducing fresh counterfactual RGB features.	scene.render_query
spatial		
$h_{t,i}^{\text{pose}}$	Relative/local candidate pose descriptor derived from a reference pose rather than raw world coordinates.	spatial.candidate_pose_feat
$h_{t,e,i}^{\text{rel}}$	Candidate-target relation descriptor in the candidate/query local frame.	spatial.candidate_target_rel_feat
$e_{a,i}^{\text{rel}}$	Query-local relative positional embedding for target, history, support, or candidate relations.	spatial.relation_rpe
model		
h_e^{tgt}	Learned selected-target token built from the actor-visible target descriptor and scene support.	model.target_token
$x_{t,i}$	Per-candidate row feature assembled from pose, relation, support, validity, provenance, and history descriptors.	model.candidate_row

1 Introduction

Active reconstruction is a coupled sensing and inference problem: the reconstruction error after a fixed acquisition budget depends as much on the selected viewpoints as on the downstream surface estimator. Classical active perception treats sensing actions as part of perception itself, and view-planning surveys formalize the generate-score-select loop for three-dimensional inspection [AWB88, Baj88, SRR03]. ARIA-NBV adopts that loop for egocentric indoor data, but asks a narrower question: when views are restricted to a finite feasible candidate table, does reconstruction-quality improvement contain planning structure beyond one-step selection?

The key empirical precedent is VIN-NBV, which replaces pure coverage with RRI, an oracle label computed from point-mesh reconstruction-error reduction after adding a query view [Fra+25]. ARIA-NBV keeps this quality-driven axis because target indoor surfaces can remain poor even when coverage proxies look saturated. The thesis transfers this idea to the Project Aria and ASE regime, where calibrated egocentric streams, trajectories, semidense points, predicted object boxes, and mesh-supervised assets support controlled oracle labels [Eng+23, Met25a, Str+24a].

The implemented seminar substrate already provides scene-level oracle RRI labels and a VINv3-style one-step candidate scorer on frozen EVL features. This thesis treats that substrate as a starting point, not the final result. The extension is target-specific: oracle target-task sampling for data generation, target-cropped labels, replayable counterfactual rollouts, and a finite-horizon value model over candidate rows. V1 actor-visible claims are evaluated through observed or predicted target descriptors and Ground-Truth Target Evaluation (GT-EVAL), not by giving GT targets to the actor.

Continuous and hierarchical NBV papers motivate later directions, but they do not define the first thesis test. GenNBV motivates continuous policy pressure, while Hestia motivates a target-then-pose hierarchy after the finite-candidate protocol is stable [Che+24, Lu+26]. Active NeRF, FisherRF, Next Best Sense,

semantic best-view, object-centric NBV, and FOVHPE motivate uncertainty, information, semantic, and target-aware diagnostics, but those channels are compared against the mesh-supervised RRI objective rather than replacing it [BLL25, Jeo+26, JLD24, LJD25, Pan+22, Str+24b].

THESIS POSITION

ARIA-NBV tests target-conditioned, quality-driven NBV on ASE/EFM3D as a finite-candidate planning problem. The core experiment first measures whether bounded oracle lookahead exposes non-myopic target-RRI headroom. If it does, a masked candidate-to-state query Q_H model is evaluated by how much of that headroom it recovers from actor-visible rollout data.

The system object is a leakage-safe finite-candidate target-specific RRI decision process on ASE/EVL. Given an ASE snippet, an oracle-sampled target task, a finite valid candidate set, and a fixed horizon, ARIA-NBV measures target-specific point-mesh RRI, estimates non-myopic headroom by oracle lookahead, and learns Q_H from non-privileged state and target-task descriptors to recover that headroom. The learned state-action value model induces the finite-action policy π_Q .

The contribution is fourfold: a leakage-safe target-specific RRI protocol, a headroom-measured finite-candidate planning test, a geometry-aware uncentred residual Q_H value model, and support-aware rollout generation. The validation sequence follows that order: target-specific RRI label construction, a calibrated target-conditioned one-step scorer over potential counterfactual views, oracle-lookahead headroom, and continuous-return residual Q_H over finite valid candidates.

Open decision: The thesis title, exact RQ wording, and final evidence scale must be re-locked once M5 evidence is known.

Source: proposal sections; advisor handout

Gate: pre-submission thesis freeze

2 Foundations

The project combines ASE scene assets, EFM3D features, oracle RRI labels, and VIN-style candidate scoring. ASE supplies Aria-like synthetic sensor trajectories, aligned semi-dense maps, and GT annotations at indoor-scene scale [Met25a]. EVL supplies frozen local voxel evidence, lifted DINO-derived features, and object-support signals for target-task descriptors and diagnostics [Met25b, Str+24a]. In the current thesis scope, EVL is local target/support evidence rather than a full long-horizon scene memory; broad scene context comes from semi-dense and fused geometry, with point-attached feature banks treated as planned representation ablations until a persisted cache and training reader exist.

2.1 Related Work

ARIA-NBV sits between classical view planning, learned reconstruction-quality scoring, and egocentric scene understanding [AWB88, Baj88]. Classical NBV work formalizes the generate-score-select loop for three-dimensional inspection and reconstruction [Ban+00, Bat+22, Bir+16], while submodular and adaptive-submodular active-sensing results explain why greedy policies can be strong when the utility has diminishing returns [GK11, KSG08, Lau+20, NWF78, SRR03]. This thesis uses that lineage as a baseline discipline rather than as a theorem claim: target-specific RRI depends on masks, object matching, candidate regeneration, occlusion, and changing support, so coverage-style approximation guarantees are treated as motivation for greedy controls, not as properties of the ARIA objective.

The closest objective precedent is VIN-NBV, which learns to rank sampled candidate views by oracle RRI instead of by pure coverage [Fra+25]. ARIA-NBV keeps that reconstruction-quality signal but changes both the state and the unit of evaluation. The actor-visible state comes from Project Aria-style egocentric streams and ASE assets [Eng+23, Met25a], and the thesis target is Target-Specific RRI (target RRI) rather than only scene-level gain. CORAL-style ordinal learning is therefore useful for a calibrated one-step scorer, but it is a control for label reliability and ranking, not the whole finite-horizon planning result [CMR19].

Project Aria, ASE, and EFM3D supply the substrate for this thesis. Project Aria defines the multimodal egocentric sensor regime and calibration assumptions, ASE provides mesh-supervised synthetic trajectories and annotations, and EFM3D/EVL contributes actor-visible local voxel evidence, lifted image features, and object hypotheses [Eng+23, Met25a, Str+24a]. The thesis does not treat EVL as a complete world model. Its role is local target/support evidence; broader scene memory is represented through semi-dense or fused point support, ray-aware occupied/free/unknown memory, and, if needed, visibility-gated point-attached feature banks.

Coverage and information-based NBV methods provide useful diagnostic channels but not the primary reward. SCONE and MACARONS estimate surface visibility or coverage from learned scene representations, while ActiveNeRF, FisherRF, and Next Best Sense use information gain and uncertainty in radiance-field or Gaussian-splatting settings [GML22, Gué+23, JLD24, Pan+22, Str+24b]. ARIA-NBV can adopt support, overlap, uncertainty, and directional-history features from this family, but the thesis compares those proxies against target RRI rather than replacing the mesh-supervised label. Object-centric, occlusion-aware, and benchmark work similarly motivates target-aware utility and semantic failure analysis, while remaining a bridge once the mesh/oracle target protocol is stable [Hu+26, Jeo+26, Ker+23, Pan+26].

Continuous-policy NBV papers define an important boundary for the thesis. GenNBV learns a generalizable continuous policy for active reconstruction, PB-NBV emphasizes efficient projection-based candidate scoring, and Hestia introduces a hierarchical target-then-pose formulation with directional observability [Che+24, Jia+25, Lu+26]. These works motivate later actor-critic, projection-shortlist, and hierarchy experiments. The core thesis remains narrower: finite candidate tables over ASE snippets, oracle target labels from GT-EVAL, and a masked Q_H model whose selected actions are re-evaluated by the same oracle.

The planned value model also borrows structure from set and geometric learning. Deep Sets gives a pooled symmetry baseline for unordered candidate or point

sets; Set Transformer supplies masked candidate-candidate interaction as an ablation; and query-centric trajectory forecasting motivates local relative positional encodings without importing a motion-forecasting decoder [Lee+19, Zah+17, Zho+23]. Geometric deep learning frames the broader design question as matching model symmetries to the output map, not making the whole system invariant [Bro+21]. For ARIA-NBV this means candidate-row permutation equivariance for Q_H scores, candidate-to-state queries before candidate-candidate self-attention, mask isolation for invalid rows, explicit target/current/candidate frames, and gravity-aware rather than full rotation-invariant features. The dedicated theory treatment in Section 2.2 turns those literature roles into the architecture ladder and acceptance tests.

Point and sparse scene encoders are best treated as representation ablations after the simpler support contract is measured. Point Transformer, PointNeXt, Point Transformer V3, KPConv, and sparse-convolutional encoders offer local geometric interaction, density handling, scalability, or sparse-volume computation [CGS19, Qia+22, Tho+19, Wu+24, Zha+21]. They become relevant if compact target, frustum, support, and lifted-feature pools leave a measurable bottleneck for target RRI ranking or finite-horizon Q_H . They are not first-line replacements for EFM3D because a thesis backbone must still provide actor-visible OBB hypotheses and reusable scene conditioning.

Finally, offline and recurrent decision-modeling papers inform the training and ablation ladder without changing the problem definition. Double DQN and dueling heads motivate selector/evaluator separation and value/advantage structure for finite candidate tables [HGS15, Wan+16]. IQL, CQL, and BCQ motivate explicit support constraints and skepticism about out-of-distribution bootstrapping, not an immediate switch away from the finite-candidate replay contract [FMP19, KNL21, Kum+20]. Trajectory Transformer, Decision Transformer, and Gumbel-Top-k provide vocabulary for sequence modeling, return conditioning, and stochastic branch generation only after the target/candidate/mask/evaluation contract is shared with the simpler baselines [Che+21, JLL21, KHW19]. Deja

View suggests a bounded weight-tied refinement ablation for candidate-context tokens, but not a claim that reconstruction-loop recurrence transfers directly to ARIA-NBV planning [Bur+26].

2.1.1 Literature Roles and Boundaries

VIN-NBV and ordinal scoring. This family provides the closest label precedent: oracle RRI, one-step candidate ranking, and CORAL-style calibration. ARIA-NBV adopts those elements as controls for target-specific label quality and scorer behavior, while treating finite-horizon lookahead as a separate question rather than stopping at myopic ranking.

Project Aria, ASE, and EVL. This stack defines the actor-visible substrate through egocentric sensing, calibration, semi-dense support, local voxel evidence, and OBB hypotheses. Its boundary is equally important: meshes and target crops remain supervision and evaluation tools, not part of the online actor-visible state.

Coverage, uncertainty, and information-gain methods. SCONE, MACARONS, FisherRF, and related radiance-field or Gaussian-splatting approaches contribute support, overlap, uncertainty, information-gain, and directional-novelty signals. In this thesis those quantities are diagnostic channels or candidate-token features, not replacements for target RRI as the primary objective.

Set and geometric models. Deep Sets, Set Transformer, and geometric-learning arguments provide the structural prior for finite candidate tables: permutation-aware scoring, masked attention, local relative encodings, and explicit symmetry tests. ARIA-NBV uses this family to shape Q_H controls before escalating to heavier point or sparse backbones.

Continuous, hierarchical, and recurrent bridges. GenNBV, Hestia, and Deja View motivate later bridge experiments in continuous control, hierarchical decomposition, or bounded refinement. They remain downstream ablations until finite-candidate labels, invalid-action masks, and replay-based evaluation are trustworthy.

2.2 Geometric Learning and Candidate-Set Theory

The geometric-learning question in ARIA-NBV is not whether a more expressive scene backbone can be added to the system. The thesis object is a typed finite-candidate decision map: for a target task, selected history, partial geometry, and an unordered set of feasible candidate poses, predict which candidate will improve target reconstruction under oracle re-evaluation. The model should therefore match the symmetries of this map before it escalates to heavier point, sparse, recurrent, or equivariant modules [Bro+21].

The first required structure is candidate-row permutation equivariance. Reordering the rows of $\mathcal{Q}_t$ may reorder per-candidate Q_H outputs, but it must not change the value assigned to the same physical candidate. Candidate-to-state queries are therefore the first context mechanism: each candidate reads the same target, map, history, budget, and local-EVL state tokens. Deep Sets supplies the minimal pooled context baseline for unordered candidate and support sets, while Set Transformer supplies masked candidate-candidate interaction once independent and candidate-to-state controls are calibrated [Lee+19, Zah+17]. The thesis should not replace the row path with only a pooled scene embedding, because the policy selects a candidate row.

The minimum algebraic contract is therefore a row-wise map over a masked set. For any permutation matrix Π acting on candidate rows, the value model should satisfy

$$f_\theta(\Pi \mathbf{X}_t, \Pi \mathbf{m}_t) = \Pi f_\theta(\mathbf{X}_t, \mathbf{m}_t)$$

where $\mathbf{X}_t = \{\mathbf{x}_{t,i}\}_{i=1}^{N_q}$ stores per-candidate actor-visible descriptors and $\mathbf{m}_t$ stores validity. Invalid rows are not low-quality actions; they are outside the admissible set. A masked scorer must therefore compute selection over

$$\mathcal{A}_t = \{i \in \{1, \dots, N_q\} : m_{t,i} = 1\}, \quad a_t^\theta = \operatorname{argmax}_{i \in \mathcal{A}_t} f_{\theta,i}(\mathbf{X}_t, \mathbf{m}_t)$$

The local-frame contract is a controlled gauge choice rather than a full invariance claim. Candidate features may use transforms such as $\mathbf{T}_{c_q}^r$, target-relative bearings, and query-local relative encodings so that a global origin change does not change

the physical row identity, while gravity alignment and camera-frustum geometry remain available task structure [Zho+19, Zho+23].

The second required structure is gauge discipline. Candidate, target, current camera, and selected-history poses should be encoded in local frames, with continuous rotation representations and query-local relative positional features where needed [Zho+19, Zho+23]. This borrows the useful part of query-centric trajectory models: local relations and relative encodings. It does not import their forecasting decoder, road-agent priors, or online streaming claims. Full global $SE(3)$ equivariance is also not the right default because the task is gravity aligned, camera-frustum limited, and tied to egocentric acquisition. Exact equivariant networks remain ablations for support encoders or candidate graphs, not the core claim [Fuc+20, SHW21].

The third required structure is directional visibility memory. A candidate view is not only a posed pinhole pose; it is a directional field from which target-local surface regions may or may not have been observed. This suggests storing selected-view history on $\mathbb{S}^2$ through a histogram, a second-moment summary, or low-order spherical harmonics before more specialized equivariant layers are introduced [e3n25]. Coverage and information-gain methods motivate support, overlap, and uncertainty features, but those features remain diagnostic channels unless they improve target-specific oracle RRI under the paired evaluation protocol [GML22, JLD24].

Object	Required structure	Testable consequence
Candidate rows	Permutation-equivariant per-row value map.	Row-shuffle tests must satisfy $f_{\theta(\Pi X, \Pi m)} = \Pi f_{\theta(X, m)}$ for every selection score.
Invalid and padded rows	Hard mask isolation.	Invalid rows cannot change valid-row scores except through explicit valid-count or support features.
Candidate-target geometry	Target-local and candidate-local relative pose features.	World-frame origin, yaw convention, and display-only camera transforms cannot become shortcuts.
Selected-view history	Directional memory on $\mathbb{S}^2$.	Novelty tests separate already-observed target directions from generic pose distance.
Actor/oracle bound-ary	Typed provenance for target descriptors, labels, and support features.	GT meshes, crops, and all-candidate renders supervise labels/evaluation only.

Table 1: Minimum geometric-learning contract for the finite-candidate value model.

This contract implies an architecture ladder with conservative attribution. The independent scorer tests whether candidate self-features already solve the target task. Candidate-to-state cross-attention tests whether fixed target, map, history, and budget tokens explain finite-horizon value without candidate-candidate coupling. A pooled DeepSets context tests whether valid-set summaries help without pairwise attention. A masked Set Transformer tests candidate-candidate interaction. QCNet-style relative positional encoding tests whether local candidate-target/history relations are the missing signal. Directional support and overlap biases test whether visibility memory or surface-support hypotheses add value. Only after these controls pass should point backbones, sparse convolutions, exact equivariant graphs, or looped refinement be treated as thesis-relevant architecture changes [Bur+26, CGS19, Qia+22, Tho+19, Wu+24].

Level	Model	Scientific role
A0	Independent candidate scorer	Locks target/candidate descriptors, CORAL calibration, and one-step ranking before context is added.
A1	Candidate-to-state query	Tests fixed target, map, history, budget, and local-EVL context while preserving a per-candidate row path.
A2	Pooled DeepSets context	Tests unordered valid-set summaries while preserving a per-candidate row path.
A3	Masked Set Transformer	Tests candidate-candidate interaction under row-shuffle and mask-isolation checks.
A4	Query-local relative encoding	Adapts QCNet-style local RPE for candidate-target, candidate-candidate, and selected-history relations.
A5	Directional/support bias	Tests visibility memory, target-frustum support, overlap, and uncertainty as features rather than proxy rewards.
A6	Uncentred residual Q_H	Tests finite-horizon recovery in continuous return units over the calibrated myopic scorer with masked Double-Q backups.
A7+	Point/sparse/exact-equivariant/recurrent encoders	Escalates only if compact descriptors bottleneck target-RRI ranking or Q_H recovery.
B0	Online candidate-table refresh	Bridge level after offline Q_H evidence: tests whether the same finite-action contract survives streaming candidate regeneration.
B1	Continuous policy / simulator controller	Post-thesis bridge only unless finite candidates fail; requires a separate action-parameterization, safety, and simulator-realism gate.

Table 2: Architecture ladder used to keep geometric-learning claims attributable.

The active method turns this theory into concrete replay fields, descriptor versions, and acceptance tests in Chapter Section 4.4. The literature review uses the same split: independent scoring and candidate-to-state queries are immediate model controls; Deep Sets, Set Transformer, and query-local relative encoding are interaction ablations; point, sparse, exact-equivariant, radiance-field, and recurrent

models are scoped representation ablations or bridge work until the finite-candidate target-RRI protocol is stable.

Validation TODO: Add measured row-shuffle, mask-isolation, duplicate-row, frame-transform, and valid-count stress-test results before claiming any geometric-learning module improves Q_H .

Source: autoresearch thesis-lit-review report; docs/contents/theory/candidate_view_dependence.qmd

Gate: architecture ablation evidence

The literature is used here to assign roles, not to broaden the thesis claim. Older active perception motivates action-conditioned sensing; VIN-NBV supplies the quality-driven candidate-ranking precedent; Project Aria, ASE, and EFM3D motivate the logged egocentric state; and offline value learning supplies replay and overestimation controls for the finite candidate table.

Role		Relevant signal	Adopt / defer
Quality-driven	NBV	Oracle RRI and ordinal one-step candidate ranking are the closest implemented precedent.	Adopt point-mesh RRI labels and a learned one-step target scorer; test whether one-step ranking is enough.
Egocentric	substrate	Logged streams and trajectory, semi-dense support, frozen EVL/EFM3D evidence, and observed/predicted OBBs form the actor-visible state.	Use observed/predicted target descriptors as actor input; keep GT geometry and crops for labels and evaluation.
Greedy sensing and finite candidates	[GK11, KSG08]	Diminishing-returns intuition; oracle-lookahead headroom as a required test.	Measure oracle-lookahead headroom before claiming a learnable non-myopic advantage.
CORAL, set models, QCNet, Double Q	[CMR19, HGS15, Lee+19, Zho+23]	Ordinal target scorer, MLP/DeepSets/Set Transformer candidate controls, query-centric relative-encoding ablation, masked fitted backups.	QCNet trajectory decoding, motion-forecasting losses, online streaming claims, IQL/CQL/BCQ, and distributional heads remain outside the initial value result.
Continuous and distance-field	NBV	Continuous policies, target-then-pose hierarchies, target/object-focused utility, and uncertainty/semantic utility channels are useful comparisons.	Use as follow-up design pressure; do not replace target RRI with coverage, uncertainty, or semantic proxy rewards.

Table 3: Source-backed literature roles for the thesis scope.

The resulting lineage is deliberately narrow:

$$\mathcal{U}_{\text{cov/unc}} \rightarrow \hat{r}_t^e(i) \rightarrow r_t^e \rightarrow G_t^{(H)} \rightarrow Q_{H,\theta}$$

Coverage and uncertainty remain diagnostics, not the thesis utility. GT meshes and GT target boxes remain oracle assets for target-task sampling, labels, and evaluation; they are not learned actor inputs unless a named privileged ablation says so. Offline and continuous RL references become meaningful only after candidate support, masks, and oracle re-evaluation are trustworthy.

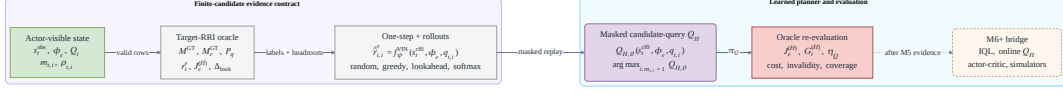


Figure 1: Evidence chain from actor-visible state and target descriptor to masked candidates, target RRI, lookahead headroom, and the Q_H model. Dashed paths are follow-up work.

Hestia informs a deferred hierarchy in which a target or look-at point is proposed before choosing a feasible pose conditioned on it [Lu+26]. In ARIA-NBV this

factorization keeps target-specific RRI as supervision/evaluation and treats feasibility projection or masks as constraints:

$$\pi_{\text{cont}}(\mathbf{x}_t, \ell_t \mid s_t, \varphi_e) = \pi_{\text{pose}}(\mathbf{x}_t \mid \ell_t, s_t, \varphi_e) \pi_{\text{look}}(\ell_t \mid s_t, \varphi_e)$$

Research TODO: Decide which bridge literature belongs in the final thesis background versus the discussion/future-work chapter.

Source: proposal related work; advisor handout

Gate: final literature pass

3 Oracle and Data Generation

This chapter defines the non-deployable machinery that turns logged ASE snippets into supervised target-conditioned NBV tasks. It owns the actor/oracle state boundary, target-task selection, candidate generation, hard validity masks, target-specific RRI labels, and rollout/replay evidence. The learned method in Chapter 4 consumes actor-visible products from this pipeline; GT meshes, GT target crops, dense candidate renders, and oracle lookahead remain label, teacher, upper-bound, or evaluation assets.

3.1 State and Visibility Boundary

The initial non-myopic experiment is a *masked finite-horizon candidate-decision process*, not a stationary deployment MDP. It is an offline, mesh-supervised *controlled counterfactual replay* process: the oracle may render selected candidates and regenerate the next candidate table as part of the experimental protocol, while the actor sees only the typed state features below.

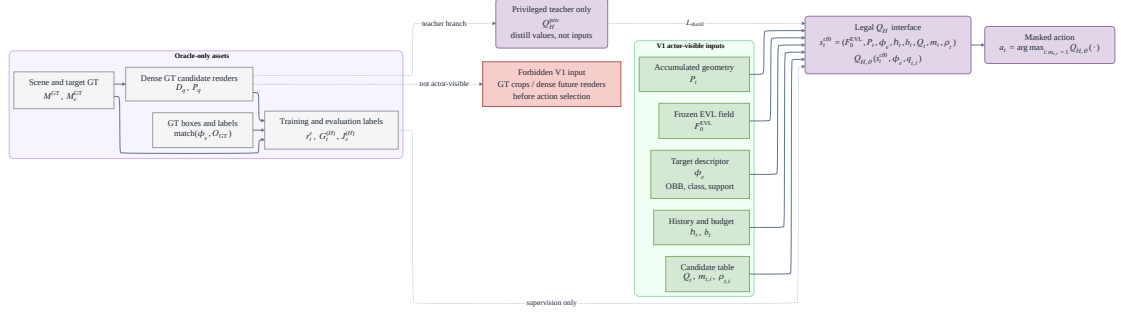


Figure 2: Actor-visible and oracle-only state boundary for the V1 thesis protocol. The learned Q_H policy may consume accumulated geometry, frozen EVL evidence, target descriptors, history, budget, candidates, masks, and invalid reasons; GT meshes, boxes, dense candidate renders, target labels, and endpoint metrics supervise labels, upper bounds, evaluation, or explicitly named teacher ablations only.

The three state spaces referenced by the protocol tuple are the codomains of the shared state equations. The logged historic state contains only the original trajectory evidence,

$$s_t^{\text{hist}} = (\mathbf{I}^{\text{rgb}}, \mathbf{X}, \mathcal{P}^{\text{semi}}, \mathbf{F}_v, e_t, b_t) \in \mathcal{S}^{\text{hist}}$$

the counterfactual actor state is the deployable state for Q_H decisions,

$$s_t^{\text{cf0}} = (\mathbf{F}_v^{\text{root}}, \mathcal{P}_t, \mathcal{Q}_t, m_{t,i}, \rho_{t,i}, e_t, b_t) \in \mathcal{S}^{\text{cf0}}$$

and the privileged oracle state is a label/evaluation state outside the actor input graph:

$$s_t^{\text{oracle}} = (s_t^{\text{cf+}}, \mathcal{M}^{\text{GT}}, \mathcal{M}_e^{\text{GT}}, \mathbf{D}_q, \mathcal{P}_q, \text{RRI}) \in \mathcal{S}^{\text{oracle}}$$

$$\mathcal{M}_{\text{NBV}} = (\mathcal{S}^{\text{hist}}, \mathcal{S}^{\text{cf0}}, \mathcal{S}^{\text{oracle}}, \{\mathcal{A}_t\}, T, r_t^e, \gamma, H)$$

The model separates logged snippet state, counterfactual actor state, and privileged oracle state because real egocentric trajectories contain modalities that are not available after synthetic view choices. The raw historic state is available on the logged ASE trajectory. The planner state keeps local root EVL evidence fixed while updating the fused geometry proxy, optional point-feature bank, selected-view history, remaining horizon metadata, target descriptor, candidates, masks, and reason codes. The privileged oracle state augments this with GT geometry, the matched target mesh, all-candidate rendered points, and oracle labels for target-task selection, label generation, upper-bound planning, and evaluation.

GT meshes, GT OBBs, GT crops, GT semantic labels, and all-candidate GT renders are oracle assets. They may define target tasks and labels, but they are not learned actor inputs unless a named upper-bound or privileged-teacher ablation explicitly says so. Invalidity is modeled as a constraint: masks apply before argmax, temperature softmax, loss targets, and bootstrap maximization. True infeasibility or absent evaluation samples are invalid rows; low immediate target support is reported as a diagnostic unless it prevents valid oracle evaluation.

3.2 Data Generation and Target-Specific RRI Labels

Target selection is part of the oracle data-generation pipeline. It samples supervised target tasks for target-conditioned NBV, rather than making a deployable claim that the learned actor discovers the target of interest on its own. The learned model receives a target descriptor and is evaluated by whether its selected views improve the matched target under oracle re-evaluation.

The target descriptor records the task and the cheap evidence available for diagnostics or later descriptor ablations:

$$\varphi_e = \text{Enc}_{\text{tgt}}(\hat{\mathbf{B}}_e, \hat{\mathbf{y}}_e, \hat{\pi}_e, A_e^{\text{proj}}, n_e^{\text{semi}}, n_e^{\text{EVL}}, \omega_e^{\text{EVL}}, \ell_e^{\text{src}}, \mathbf{T}_{r_t,e}, \mathbf{T}_{c_t,e})$$

In this descriptor, $\hat{\mathbf{B}}_e$ is observed, predicted, or otherwise proposed OBB geometry for the target task; $\hat{\mathbf{y}}_e$ is class probabilities or class embedding; $\hat{\pi}_e$ is confidence; A_e^{proj} is projected area; n_e^{semi} and n_e^{EVL} are semidense and EVL support counts; ω_e^{EVL} records local EVL coverage; ℓ_e^{src} records the actor-visible source mode; and $\mathbf{T}_{r_t,e} / \mathbf{T}_{c_t,e}$ record reference- and current-frame target geometry. These fields are not all target-task gates in the first implementation. Class, confidence, current projection, semidense support, EVL support, distance, and target bearing are retained as descriptor and audit fields so that later subsets can ask whether the target-conditioned model depends on semantic correctness or observation quality.

3.2.1 Seminar Oracle Substrate and Thesis Delta

The seminar paper is implemented evidence for the one-step scene-level oracle substrate, not the current thesis objective. Its labeler constructs a candidate table for an ASE snippet, renders candidate depth from the GT mesh under a calibrated camera/rasterization convention, backprojects and fuses candidate points with the current semi-dense reconstruction, and computes point-mesh accuracy/completeness as RRI labels [Fra+25, Met25c]. ARIA-NBV reuses that substrate for label provenance, but changes the task unit: the thesis sampler first creates target tasks, target crops define the error surface, and selected counterfactual transitions are written to a standalone rollout store for finite-horizon Q_H .

Seminar material	Thesis adaptation	Placement
GT-mesh depth rendering and point-mesh RRI	Reused as label provenance; target crops replace whole-scene error as the primary objective.	Main data-generation method plus appendix camera-convention details.
Legacy shell candidate sampling	Kept as a historical/free-shell abstraction; current target-conditioned mixtures own the main candidate distribution.	Candidate/replay method and appendix, not a default-policy claim.
CORAL ordinal scorer and bin diagnostics	Reused for myopic target-scorer calibration and expected-RRI controls before residual Q_H .	Value-model method and evaluation evidence gates.
Immutable one-step VIN offline store	Kept separate from selected-transition rollouts.zarr; all-candidate labels train the scorer, selected transitions train bootstrapped Q_H .	Data-flow appendix and replay evidence matrix.
Run-specific W&B and cache-size notes	Historical diagnostics only until regenerated from current manifests and configs.	Appendix/open-work TODO, not final experiment evidence.

Table 4: How seminar-paper implementation evidence is adapted without drifting into thesis claims.

Conflict: Do not copy seminar scene-level RRI, legacy shell sampler, one-step VIN store, or run-specific CORAL/W&B results as current target-conditioned finite-horizon evidence. They are implementation substrate or historical diagnostics unless regenerated under the target-task and rollout-store protocol.

Source: docs/typst/seminar_paper/main.typ; docs/typst/seminar_paper/sections/05-oracle-rrr.typ; docs/typst/seminar_paper/sections/07-training-objective.typ; docs/typst/seminar_paper/sections/12h-appendix-offline-cache.typ

Gate: final data-generation appendix and experiment-manifest refresh

3.2.2 Target Selection

Automatic target selection constitutes the first procedural layer in target-centric oracle data generation. Given a historic snippet and its egocentric encodings, the sampler chooses target tasks for which supervised target-conditioned NBV is meaningful. A useful target task must be identifiable, evaluable, and potentially action-sensitive: the oracle must know which GT object the task refers to, target-specific error must be computable, and at least some feasible candidate views

should expose non-marginal target-specific RRI after oracle evaluation. Near-solved targets are therefore not discarded before storage, but their low headroom must be measured and preserved as evidence.

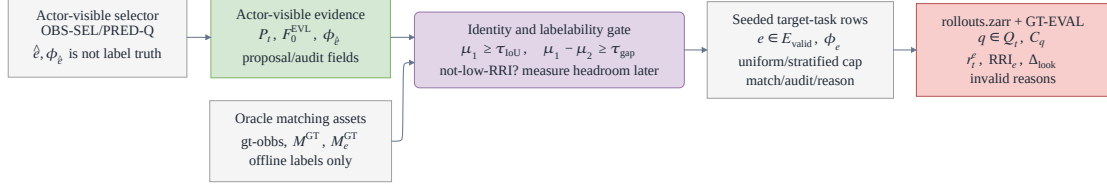


Figure 3: Oracle target-task sampling contract. GT OBBs and meshes define identity-valid supervised target tasks through an IoU and ambiguity-gap gate, a deterministic capped sampler writes target-task rows with descriptor and audit fields, and rollout generation later measures target RRI and headroom. The actor-visible target selector remains a separate diagnostic or later deployment contract, not the source of thesis labels.

The cheap admission gate is identity matching. A proposed target OBB is identity-valid when it matches exactly one GT target OBB by configured 3D IoU and ambiguity margin:

$$\mu_{\text{id}}(\hat{e}, e) = \text{IoU}_{3\text{D}}(\hat{\mathbf{B}}_{\hat{e}}, \mathbf{B}_e)$$

$$a_{\text{id}}(\hat{e}) = 1 \text{ iff } \begin{cases} \mu_1 \geq \tau_{\text{IoU}} \\ \mu_1 - \mu_2 \geq \tau_{\text{gap}} \end{cases}$$

Here μ_1 and μ_2 are the best and second-best target-to-GT IoU scores for the proposal. The threshold values are protocol parameters, not theory constants. The implementation should use a moderate default and report a threshold sweep so that coverage can be inspected under looser and stricter identity definitions.

Class prediction and confidence do not decide target-task eligibility in the first pass. They are recorded because semantic correctness may matter for target descriptors and failure analysis, but the oracle can compute target-specific RRI for a geometrically identified target even when the class head is noisy. Current RGB projection, semidense support, and EVL support are likewise audit fields rather than hard gates unless a later subset explicitly asks for projected-only or support-qualified targets.

The sampler keeps rollout cost bounded by selecting a capped number of identity-valid targets per source snippet. Within that cap, it samples uniformly with a deterministic seed rather than always taking the largest or highest-IoU object. The audit surface must still report how many identity-valid targets existed before the cap, how many were selected, and how their IoU, ambiguity gap, class, confidence, projected area, support, distance, and bearing were distributed.

Headroom is evaluated after target selection because it depends on oracle candidate scoring. The rollout store should persist identity-valid targets even when measured target headroom is low; training and evaluation loaders can then filter or stratify by headroom band. This preserves negative evidence: if many well-identified targets have no candidate with useful gain, the result is a candidate/

support/headroom limitation rather than a target-selection failure hidden by pre-filtering.

Counterfactual trajectory naturalness is a candidate and rollout diagnostic, not an identity gate. Hard turns, target-bearing changes, and support collapse should be measured with candidate provenance and invalid-reason fields so that the thesis can distinguish target identity failures from unrealistic or unsupported view proposals.

3.2.3 Candidate View Generation

Candidate view generation is the second procedural layer: it turns one target task into a finite action table for one rollout state. The current thesis profile deliberately uses a small family mixture rather than the older unconstrained shell from the seminar paper. The checked-in data-generation profile consumes the strict offline actor-state store, uses a training split for the first real audit subset, samples one oracle target task per source sample, and writes a separate rollout/replay store. It is an audit-scale thesis default, not an LRZ path template and not final scale evidence.

At rollout step t , candidate generation constructs a full shell

$$\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_q}, \quad N_q = 60$$

with one fixed provenance component $k(i)$ per row. The canonical `v1_realistic_3family` mixture is

Component	Rows	Center family	View family
<code>forward_local</code>	24	<code>forward_local</code>	<code>forward_rig</code>
<code>target_bearing_local</code>	24	<code>target_bearing_local</code>	<code>target_point</code>
<code>lateral_target_bypass</code>	12	<code>lateral_target_bypass</code>	<code>target_point</code>

Table 5: Canonical three-family finite candidate table for real thesis rollouts. Counts are full-shell rows; valid-action counts are measured after geometry and motion pruning.

For each row, the raw direction is sampled in the reference rig frame. The realistic profile uses the forward-biased Power Spherical distribution from the current implementation [DAT20]:

$$\mathbf{u}_i \sim \text{PS}(\mathbf{e}_z, \kappa), \quad p(\mathbf{u}) = c_\kappa (1 + \mathbf{e}_z^T \mathbf{u})^\kappa, \quad \kappa = 8$$

The draw is mapped into configured azimuth and elevation caps without rejection.

With $\psi = \text{atan2}(u_x, u_z)$ and $u_y = \sin \theta$, the cap transform is

$$\psi' = \psi \frac{\Delta_\psi}{2\pi}, \quad y' = \sin \theta_{\min} + \frac{u_y + 1}{2} \cdot (\sin \theta_{\max} - \sin \theta_{\min})$$

$$\mathbf{d}_i^0 = \left\| \left(\sqrt{1 - y'^2} \sin \psi', y', \sqrt{1 - y'^2} \cos \psi' \right) \right\|$$

The three position families then reinterpret this capped direction. Let $\mathbf{f} = \mathbf{e}_z$ be the rig-forward unit vector, $\mathbf{b}_e$ the actor-visible target bearing in the reference frame, $\mathbf{l}_e = \|\mathbf{e}_y \times \mathbf{b}_e\|$ the horizontal lateral direction, and $\mathbf{e}_y$ the world-up direction expressed in the sampling frame. The family directions are:

$$\mathbf{d}_i^{\text{forward}} = \left\| \mathbf{f} + \alpha_f (\mathbf{d}_i^0 - (\mathbf{d}_i^0 \cdot \mathbf{f}) \mathbf{f}) \right\|, \quad \alpha_f = 0.45$$

$$\mathbf{d}_i^{\text{target}} = \left\| \mathbf{b}_e + \alpha_t (\mathbf{d}_i^0 - (\mathbf{d}_i^0 \cdot \mathbf{b}_e) \mathbf{b}_e) \right\|, \quad \alpha_t = 0.4$$

$$\mathbf{d}_i^{\text{bypass}} = \left\| 0.55 \mathbf{b}_e + 0.85 \text{sign}(d_{i,x}^0) \mathbf{l}_e + \text{clip}(d_{i,y}^0, -0.35, 0.35) \mathbf{e}_y \right\|$$

Finally, the sampler draws a radius and transforms the reference-frame offset into world coordinates:

$$r_i \sim \mathcal{U}(0.25, 1.1), \quad \mathbf{c}_i^w = \mathbf{T}_r^w \left(r_i \mathbf{d}_i^{k(i)} \right)$$

`forward_local` keeps the reference rig orientation. The two target-looking families orient the camera to the selected actor-visible target center $\mathbf{p}_e$:

$$\mathbf{z}_i^w = \|\mathbf{p}_e - \mathbf{c}_i^w\|, \quad \mathbf{y}_i^w = \|\mathbf{e}_y - (\mathbf{e}_y^T \mathbf{z}_i^w) \mathbf{z}_i^w\|, \quad \mathbf{x}_i^w = \mathbf{y}_i^w \times \mathbf{z}_i^w$$

These equations are the mathematical description of the implemented sampler, not a claim that the mixture is optimal. The three-family design has a direct downstream impact: `forward_local` preserves egocentric motion continuity; `target_bearing_local` tests whether moving along the target bearing produces supervised target gain; and `lateral_target_bypass` creates side-step views that may improve occluded target surfaces without leaving the local walking envelope. If the target-aware families do not survive pruning, the resulting rollout dataset

degenerates into a forward-only dataset and cannot support a target-conditioned planning claim.

Pruning converts the full shell into a compact valid-action table. A row remains valid only if it lies in the snippet occupancy support, stays clear of the GT mesh, avoids straight-line path collision, and satisfies local egocentric motion limits:

$$\|\mathbf{o}_i\|_2 \leq 1.0 \text{ m}, \quad |\Delta h_i| \leq 0.25 \text{ m}, \quad \max(0, -o_{i,z}) \leq 0.25 \text{ m}, \quad \Delta\psi_i \leq 70 \text{ deg}$$

The full shell is still retained with `position_id`, `strategy_id`, `mixture_id`, `sampler_probability`, rule masks, debug diagnostics, and invalid-reason bitsets. Invalid candidates are hard-masked constraints with explicit reasons. They are never low-RRR examples, and they must not enter Q_H argmax, softmax, or loss targets. The canonical real config requires at least 15 valid root actions for a 60-row shell, matching the first production gate:

$$N_{\text{valid}} \geq \max(12, \text{ceil}(0.25N_q))$$

This threshold is a data-support guard: it prevents low-support roots from masquerading as planning evidence, while preflight still reports the blocked roots and per-family failure modes.

3.2.4 Rollout Branch Sampling and Dataset Impact

Rollout generation samples finite branches over the valid candidate table. The canonical thesis profile materializes four recipe families:

Recipe	H	B	Beam	Selection rule
<code>random_valid</code>	1	1	1	Uniform over valid rows.
<code>oracle_greedy</code>	1	1	1	Argmax of oracle target-root gain.
<code>oracle_lookahead</code>	2	2	2	Bounded oracle greedy branches.
<code>temperature_softmax</code>	2	2	2	Softmax over robust oracle scores with $\tau = 1$.

Table 6: Canonical rollout recipes for the real thesis profile. The recipes create replay diversity and oracle-lookahead references; they do not train Q_H by themselves.

For stochastic branches, let s_i be the finite oracle score of valid row i . The robust logit used for temperature-softmax is

$$\ell_i = \frac{s_i - \text{median}(s)}{\text{IQR}(s)\tau}, \quad P(i \mid m_i = 1) = \frac{\exp(\ell_i)}{\sum_{j:m_j=1} \exp(\ell_j)}$$

The downstream effect of these choices is scientific rather than cosmetic. The target source controls what counts as a supervised task; the candidate mixture controls the support on which Q_H can learn finite-action values; the validity rules decide which actions are admissible; and the branch sampler determines whether the replay store contains only myopic winners or also valid lower-ranked alternatives. A trustworthy thesis dataset must therefore report, per scene and target, selected target counts, valid candidates, valid candidates by `position_id`, invalid reasons by family, selected-family histograms, marginal and cumulative target-root gain, diagnostic state-relative target RRI, and storage/retention settings. Only after these diagnostics show non-degenerate target-aware support should failures or successes be attributed to planning rather than to the data-generation profile.

3.2.5 Target-Specific RRI

Let $C_e(\mathcal{P}_t)$ denote the oracle-only crop of accumulated points to the matched target region. The target error is the target-cropped version of the VIN-NBV RRI objective [Fra+25]: point-to-mesh accuracy plus mesh-to-point completeness on the crop.

$$\Delta_t^e = d(C_e(\mathcal{P}_t), \mathcal{M}_e^{\text{GT}}) = D_{P \rightarrow M, t}^e + D_{M \rightarrow P, t}^e$$

Area weighting or uniformly sampled target-surface points prevent target-specific RRI from reflecting mesh tessellation density. For reproducibility, the current implementation computes the point-mesh distances through `OracleRRI.score` and `chamfer_point_mesh_batched`, while `aria_nbv.pose_generation.target_counterfactuals` owns the target crop. Empty or unsupported target crops are invalid label cases, not low-RRI samples.

The immediate training reward adapts VIN-NBV’s reconstruction-improvement idea to a target crop and normalizes by the root target error rather than

the current error [Fra+25]. This makes equal-horizon rollouts additive against a common root baseline:

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\Delta_0^e + \varepsilon}$$

The finite-horizon return is the discounted sum of those target rewards along a selected counterfactual branch. It is a training target for Q_H , not a claim that the deployed system has an online continuous-control policy:

$$G_t^{(H)} = \sum_{k=0}^{H-1} \gamma^k r_{t+k}^e$$

Endpoint gain is the primary fixed-budget comparison metric because it measures the target quality after the same number of acquisitions for each policy:

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

The log-gain variant is retained as a scale-sensitivity ablation, mirroring the broader NBV literature’s use of logarithmic error reduction while keeping the root-normalized endpoint gain as the default thesis metric:

$$J_{e,\log}^{(H)} = \log(\Delta_0^e + \varepsilon) - \log(\Delta_H^e + \varepsilon)$$

$J_e^{(H)}$ is the primary fixed-horizon endpoint metric, $G_t^{(H)}$ is the rollout training return, and $J_{e,\log}^{(H)}$ is only an algebraic scale-sensitivity ablation. The default immediate rollout / Q_H reward is root-normalized by Δ_0^e , so with $\gamma = 1$ its additive return telescopes to endpoint gain up to epsilon under equal-horizon, equal-budget comparisons. The current-error denominator defines state-relative one-step RRI for diagnostics and VIN compatibility only; the final discount and clipping policy remain open protocol parameters.

Open decision: Lock identity-IoU thresholds, ambiguity gap, per-snippet target cap, near-solved-target filtering policy for loaders, clipping, and final gamma policy.

Source: target-selection interview; target-selection autoresearch report

Gate: RQ1/RQ2 protocol freeze

Validation TODO: Regenerate current manifest-backed counts for one-step labels, target-task samples, rollout rows, storage footprint, render cost, invalid reasons, and target-headroom strata before presenting scale or runtime claims.

Source: seminar offline-cache appendix; current rollout-store contract

Gate: M1/M2 data-generation evidence

4 Method

ARIA-NBV is a target-conditioned, finite-candidate NBV problem in the Project Aria / ASE / EFM3D observation regime. The learned method tests whether bounded planning over a valid finite candidate table improves target reconstruction quality beyond myopic selection. It does not claim first-order continuous control, complete scene reconstruction, or fresh counterfactual image understanding.

4.1 Scene-Representation Requirements

The planned value model needs more than a generic 3D feature extractor. A thesis-grade perception stack must provide actor-visible target hypotheses, especially OBB detections, while the scene memory must preserve reusable geometry, uncertainty, free/unknown evidence, and history for target-, history-, and candidate-conditioned Q_H queries. These are separate interfaces: a detector can propose the target without being the scene memory, and a scene memory can support candidate scoring without predicting OBBs. The current default therefore remains EFM3D / EVL for Aria-native target and local evidence: it is trained for the Project Aria / ASE regime, lifts multi-frame image features into a gravity-aligned 3D volume, uses semi-dense support, and exposes OBB/object evidence [Met25b, Str+24a]. A replacement perception backbone is only thesis-relevant if it preserves this target-detection contract; a replacement memory is relevant if it improves candidate-conditioned evidence without breaking actor-visible provenance.

The central asymmetry is that the historic state is rich and multimodal, while counterfactual states are not. Logged snippets contain calibrated RGB/SLAM streams, poses, semidense points, EVL evidence, and observed or predicted OBBs. Counterfactual successors can reliably add selected geometry, support counts, visibility history, oracle-rendered labels, and actor-visible summaries derived from those sources; they cannot assume fresh RGB, DINO, semantic, or detector outputs at unvisited candidate poses unless a separate renderable or learned modality generator is introduced and validated. This prevents a model from winning by

consuming modalities that exist only for the logged trajectory or only inside the oracle.

The important failure mode is support mismatch. The EVL voxel field is rooted in the snippet evidence and has finite local support; a target OBB, target-local surface cell, or candidate frustum can be physically relevant while partially or fully outside that field. For Q_H , an EVL read therefore contributes both a feature and a coverage diagnostic, not a proof that the target is irrelevant:

$$\omega_{t,i}^{\text{EVL}} = \frac{1}{K} \sum_{k=1}^K \mathbb{1}[x_{t,i,k} \in \mathcal{V}_0^{\text{EVL}}], \quad \mathbf{g}_{t,i}^{\text{EVL}} = \text{Pool}(\{\mathbf{F}_0^{\text{EVL}}(x_{t,i,k}) : x_{t,i,k} \in \mathcal{V}_0^{\text{EVL}}\})$$

Here $x_{t,i,k}$ are target-crop, frustum, or target-frustum-intersection query points for candidate i , $\mathcal{V}_0^{\text{EVL}}$ is the root EVL support volume, $\omega_{t,i}^{\text{EVL}}$ is the actor-visible local-support fraction, and $\mathbf{g}_{t,i}^{\text{EVL}}$ is a pooled local evidence token. Low $\omega_{t,i}^{\text{EVL}}$ should usually be a support feature, uncertainty feature, or preflight warning; it becomes hard invalidity only when the row cannot be evaluated or cannot produce a meaningful actor-state transition.

This is why the final EVL head fields are a baseline, not the thesis memory optimum. Fields such as `occ_pr`, `cent_pr`, `bbox_pr`, and `clas_pr` are compact actor-visible predictions, but they are local, task-collapsed, and tied to the root snippet. They do not by themselves preserve broad free/unknown space, target-candidate visibility, directional history, or logged appearance ambiguity. Enlarging the EVL cube is therefore a useful ablation for local-support failure, not the default state design: dense 3D cost grows with volume, and a larger root cube still cannot create counterfactual RGB, DINO, detector, or history evidence. The tractable optimum is local EVL evidence plus sparse ray-aware actor-visible memory, with compressed logged descriptors only after visibility and provenance are enforced.

Requirement		Reason	Thesis consequence
Target-proposal interface	in-	Target-specific RRI and target-conditioned Q_H need a selectable entity hypothesis, not only a scene embedding.	Models without observed/predicted OBB support are scene-memory or feature ablations unless another actor-visible target proposer supplies the target record.
Aria / ASE domain fit		Egocentric cameras, gravity alignment, semidense support, synthetic-to-real gaps, and ASE OBB taxonomy are part of the data contract.	Prefer EFM3D / EVL unless another model is trained or adapted on ARIA/ASE-style data and reports OBB quality.
Reusable conditioning state	condition-	Q_H must condition on target, selected history, support, visibility, free/unknown evidence, and candidate geometry across many candidate rows.	Expose typed query pools and ray-aware candidate queries rather than one final-pose voxel tensor: target-support pool, candidate support, candidate render/query, and directional-history support.
Actor-visible provenance	prove-	Historic and counterfactual states have different modality availability.	Every token records source role, frame, feature model, support count, uncertainty, and whether it came from logged observation, selected successor depth, or oracle-only evaluation.
Controlled scale		The thesis should not become a large foundation-model replacement project.	A smaller or cheaper encoder is desirable, but only after OBB quality, queryability, storage, runtime, and invariance tests pass.

Table 7: Hard requirements for a backbone or scene encoder to enter the thesis-core Q_H path. This makes EVL an anchor, not a ceiling. The near-term representation question is whether the current final-pose voxel field hides useful evidence. The first ablations should keep the EFM3D/EVL target detector and add better actor-visible conditioning: semidense/fused point support, a sparse ray-aware occupied/free/unknown memory, EVL internal or crop reads around target hypotheses, selected-

depth successor summaries, explicit directional visibility memory, and only then compressed DINO-on-point features lifted from logged views. These products can give Q_H a broader queryable state without claiming that a new model has solved Aria-native OBB detection.

The logged DINO-on-point ablation uses the same geometric contract as EFM3D lifting, but the carrier is a semidense or fused world point rather than a local voxel center:

$$\mathbf{p}_{j,c,\tau} = \mathbf{T}_w^{c,\tau} \mathbf{p}_j, \quad (u_{j,\tau}, v_{j,\tau}, \alpha_{j,\tau}) = \pi_{\kappa_\tau}(\mathbf{p}_{j,c,\tau})$$

The descriptor is sampled only from logged feature maps and pooled through visibility-gated, actor-visible support weights:

$$\mathbf{f}_{j,\tau} = \text{Sample}(\mathbf{F}_\tau^{2D}, u_{j,\tau}, v_{j,\tau})$$

$$m_{j,\tau}^{\text{vis}} = \alpha_{j,\tau} m_{j,\tau}^{\text{obs/depth}} m_{j,\tau}^{\text{quality}}$$

$$w_{j,\tau} = m_{j,\tau}^{\text{vis}} q_j r_{j,\tau}, \quad \bar{\mathbf{f}}_j = \frac{\sum_\tau w_{j,\tau} \mathbf{f}_{j,\tau}}{\sum_\tau w_{j,\tau} + \varepsilon}$$

$$\mathbf{f}_j^{\text{DINO-comp}} = \text{Compress}(\bar{\mathbf{f}}_j), \quad n_j^{\text{valid}} = \sum_\tau m_{j,\tau}^{\text{vis}}$$

Here $\alpha_{j,\tau}$ is only the calibrated projection-valid mask. The visibility gate $m_{j,\tau}^{\text{vis}}$ must additionally come from native semidense observation lineage, depth consistency, a conservative z-buffer, or a quality filter; projection validity alone is not evidence that the point was visible. This path can cover observed points outside the root EVL voxel cube, but it does not create fresh RGB, DINO, detector, or EVL evidence for unvisited future candidate poses.

The representation ladder is therefore short. EFM3D/EVL is the anchor because it supplies Aria-native OBB and support evidence. Semidense or fused observations and a sparse ray-aware map are the first broad-memory ablation because they can preserve observed surface, observed free space, unknown space, support counts, and directional history outside the root EVL cube while remaining actor-visible. EVL internal or crop reads test whether final heads hide local target

evidence. Visibility-gated logged DINO descriptors then test appearance memory. Point, sparse, renderable, and recurrent encoders are later support-memory ablations; Cube R-CNN-style features are detector or ROI diagnostics unless they match the EFM3D target-evidence role on ARIA/ASE data [Bra+23, Bur+26, Ker+23, Qia+22, Tho+19, Wu+24].

4.2 Descriptor and Encoding Protocol

This subsection defines the actor-visible descriptor protocol consumed by Q_H . It is intentionally narrower than the rollout store: replay rows persist identifiers, canonical poses, masks, selected transitions, successor references, oracle labels, and provenance, while the training reader derives versioned descriptor tensors from those facts. The notation separates records, learned tokens, spatial relations, support reads, and provenance so that the model cannot silently consume oracle-only information or arbitrary coordinate-system shortcuts. The derived input bundle is

$$\left(\mathbf{h}_e^{\text{tgt}}, \Phi_t^{\text{scene}}, \mathbf{H}_t, \{ \mathbf{x}_{t,i}, \mathbf{e}_{a,i}^{\text{rel}}, \mathbf{m}_{t,i}, \boldsymbol{\rho}_{t,i} \}_{i=1}^{N_q} \right)$$

Here $\mathbf{h}_e^{\text{tgt}}$ is the selected-target token, Φ_t^{scene} is queryable actor-visible scene memory, $\mathbf{H}_t$ is selected-history state, and each candidate row carries typed self, relation, support, validity, and provenance descriptors. Hidden GT target crops, GT matches, target errors, and oracle returns are label or evaluation products only; they are never part of the V1 actor input.

The target descriptor is the first descriptor because it fixes which object the value model is allowed to care about. It separates identity, actor-visible support, and provenance from later GT association:

$$\boldsymbol{\varphi}_e = \text{Enc}_{\text{tgt}} \left(\hat{\mathbf{B}}_e, \hat{\mathbf{y}}_e, \hat{\pi}_e, A_e^{\text{proj}}, n_e^{\text{semi}}, n_e^{\text{EVL}}, \omega_e^{\text{EVL}}, \ell_e^{\text{src}}, \mathbf{T}_{r_t,e}, \mathbf{T}_{c_t,e} \right)$$

In this equation the vector $\boldsymbol{\varphi}_e$ denotes the actor-visible target representation, while Enc_{tgt} is the constructor. This avoids the older ambiguity where φ_{target} named both an encoder and a target-like representation. Confidence is written $\hat{\pi}_e$ rather

than $\hat{p}_e$ to keep probabilities distinct from points or positions, and source mode is written ℓ_e^{src} rather than a state-like s .

The learned target token then augments the descriptor with support read from scene memory:

$$\mathbf{h}_e^{\text{tgt}} = \text{MLP}_{\text{tgt}}(\text{concat}(\boldsymbol{\varphi}_e, \mathbf{g}_e^{\text{tgt}}))$$

The target fields include observed or predicted OBB geometry, class/confidence, projected area, semidense and EVL support, local EVL coverage, source mode, and root/current-relative target geometry. Low target support is therefore observable evidence and a reporting stratum. It becomes hard invalidity only through the target-task protocol when the target cannot be matched, cropped, or evaluated.

The scene descriptor is a queryable support memory, not one dense final-pose tensor:

$$\boldsymbol{\Phi}_t^{\text{scene}} = (\mathbf{M}_t^{\text{ray}}, \mathbf{X}_t^{\text{pt}}, \mathbf{F}_t^{\text{DINO@pt}}, \mathbf{E}_0^{\text{EVL-local}}, \mathbf{O}_t^{\text{pred}}, \mathbf{M}^{\text{dir}})$$

This composite keeps sparse ray-aware occupied/free/unknown memory, semi-dense or fused point tokens, optional compressed logged DINO descriptors, root-local EVL evidence, predicted target hypotheses, and directional-history summaries as distinct channels. The split matters because logged history may contain calibrated RGB/DINO and semidense evidence, while counterfactual successor rows may add only selected geometry, ray/free-space updates, support counts, and missing-visual-descriptor masks.

Optional logged appearance enters through point-attached tokens only after projection, visibility, compression, and source provenance are explicit:

$$\mathbf{x}_j^{\text{pt}} = \text{concat}(\mathbf{p}_j, \mathbf{f}_j^{\text{DINO-comp}}, \sigma_j^{-1}, n_j, \mathbf{a}_j^{\text{hist}})$$

This planned DINO-on-point branch can represent observed points outside the root EVL voxel cube, but it does not create fresh RGB, DINO, detector, or EVL evidence at unvisited candidate poses.

Candidate pose is only the candidate self descriptor, not the whole row. Canonical poses remain stored as rigid transforms; model readers derive a relative pose feature from a reference pose r_t :

$$\mathbf{h}_{t,i}^{\text{pose}}(q_{t,i}; r_t) = \text{concat}\left(\xi_{r_t,i}^{\text{rel}}, \mathbf{R}_{r_t,i}^{6\text{D}}, \Delta h_{t,i}, \mathbf{u}_{t,i}^{\text{up/frustum}}\right)$$

For logged states, r_t is the current/root actor pose used to define the decision state; for counterfactual successors, it is the preceding selected pose. This reference pose is a gauge choice, not another observation modality. It lets the model see motion feasibility and egocentric geometry without depending on arbitrary world origin or yaw. Continuous 6D rotations remain a stable neural representation for rotations [Zho+19], and learnable Fourier features remain a useful scalar/vector encoding control [Li+21], but they should be applied to relative transforms and physically meaningful scalars rather than raw world pose by default.

The candidate-target relation is a separate descriptor:

$$\mathbf{h}_{t,e,i}^{\text{rel}}(q_{t,i}, e) = \text{concat}\left(\mathbf{R}_{t,i}^\top(\mathbf{c}_e - \mathbf{c}_{t,i}), \|\mathbf{c}_e - \mathbf{c}_{t,i}\|_2, \cos \theta_{t,e,i}^{\text{opt}}, \beta_{t,e,i}^{\text{elev}}, \lambda_{t,e,i}^{\text{obb}}\right)$$

This relation uses candidate-local target displacement, range, bearing or optical-axis alignment, elevation, and OBB/frustum overlap. It replaces the earlier overloaded pose vector that combined absolute translation, 6D rotation, target distance, angle, overlap, and sampler provenance. Sampler family and source lineage remain useful diagnostics, but they are provenance features, not geometric pose.

Candidate support is then queried from the scene memory in three typed pools:

$$\begin{aligned}\mathbf{g}_e^{\text{tgt}} &= \text{Pool}_{\mathbf{p}_j \in \hat{\mathbf{B}}_e} \mathbf{x}_j^{\text{pt}} \\ \mathbf{g}_{t,i}^{\text{fr}} &= \text{Pool}_{\mathbf{p}_j \in \text{Frustum}(q_{t,i})} \mathbf{x}_j^{\text{pt}} \\ \mathbf{g}_{t,e,i}^{\text{cap}} &= \text{Pool}_{\mathbf{p}_j \in \hat{\mathbf{B}}_e \cap \text{Frustum}(q_{t,i})} \mathbf{x}_j^{\text{pt}}\end{aligned}$$

The pools summarize selected-target support, candidate-frustum support, and the target-frustum intersection. They are cheap support descriptors, not visibility

truth by themselves. The candidate observation branch also needs a ray-aware query over occupied, free, unknown, hit, target, support, uncertainty, and directional-novelty channels:

$$\mathbf{g}_{t,i}^{\text{ray}} = \text{RenderQuery}(\mathbf{M}_t^{\text{ray}}, q_{t,i}, \hat{\mathbf{B}}_e) = (\mathbf{D}_{t,i}^{\text{near}}, \mathbf{L}_{t,i}^{\text{free}}, \mathbf{L}_{t,i}^{\text{unk}}, \mathbf{M}_{t,i}^{\text{hit}}, \mathbf{W}_{t,i}^{\text{target}}, \mathbf{C}_{t,i}^{\text{support}}, \boldsymbol{\Sigma}_{t,i}^{\text{geom}}, \nu_{t,i}^{\text{dir}})$$

Query-centric relation encodings are introduced only after the self and support descriptors exist. The transferable part of QCNet is the local coordinate-system and relative positional-embedding discipline: key/value elements are described relative to the query element, giving useful invariances without relying on global scene coordinates [Zho+23]. In ARIA-NBV, each candidate can query target, history, support, and other candidate tokens through features expressed in the query candidate’s local frame:

$$\begin{aligned} \boldsymbol{\delta}_{a|i}^{\text{p}} &= \mathbf{R}_{t,i}^\top (\mathbf{p}_a - \mathbf{c}_{t,i}), \quad \boldsymbol{\delta}_{a|i}^{\text{R}} = \mathbf{R}_{t,i}^\top \mathbf{R}_a \\ \boldsymbol{\eta}_{a|i} &= \text{concat}(\boldsymbol{\delta}_{a|i}^{\text{p}}, \|\boldsymbol{\delta}_{a|i}^{\text{p}}\|_2, \text{enc}_R(\boldsymbol{\delta}_{a|i}^{\text{R}})), \quad \mathbf{e}_{a,i}^{\text{rel}} = \psi_{\text{rel}}(\mathcal{F}(\boldsymbol{\eta}_{a|i})), \quad a \in \{j, e, k\} \end{aligned}$$

This imports the geometrically useful idea, not QCNet’s trajectory decoder, anchor losses, road-agent metrics, or streaming prediction claim. The default first scorer can ignore these relation embeddings; Deep Sets and Set Transformer controls should establish row-order and mask behavior before relation-biased candidate attention is promoted [Lee+19, Zah+17].

Selected-history visibility is a different descriptor family from pose. Candidate orientation uses continuous 6D rotation features [Zho+19], but accumulated observation history lives on $\mathbb{S}^2$. For a target-local point or voxel center $\mathbf{v}$ and a previously selected camera center $\mathbf{c}_k$, the observed direction is

$$\mathbf{d}_k(\mathbf{v}) = \frac{\mathbf{c}_k - \mathbf{v}}{\|\mathbf{c}_k - \mathbf{v}\|_2}$$

The planned actor-visible branch stores this history either as low-order spherical-harmonic coefficients [e3n25]

$$\mathbf{h}^{\text{dir}}(\mathbf{v}) = \sum_{k < t} w_k(\mathbf{v}) \mathbf{Y}_L(\mathbf{d}_k(\mathbf{v}))$$

or, as the thesis-safe default, as a second-moment summary over unit directions,

$$\mathbf{M}^{\text{dir}}(\mathbf{v}) = \sum_{k < t} w_k(\mathbf{v}) \mathbf{d}_k(\mathbf{v}) (\mathbf{d}_k(\mathbf{v}))^\top$$

from which the candidate can read a directional novelty score:

$$\nu_{t,i}^{\text{dir}}(\mathbf{v}) = 1 - \frac{(\mathbf{d}_{t,i}(\mathbf{v}))^\top \mathbf{M}^{\text{dir}}(\mathbf{v}) \mathbf{d}_{t,i}(\mathbf{v})}{\text{tr}(\mathbf{M}^{\text{dir}}(\mathbf{v})) + \varepsilon}$$

The directional-memory diagram in Figure 4 keeps this branch separate from generic pose tokens: selected views first accumulate target-local directional evidence, and each valid candidate then queries whether it sees the target from a genuinely new direction.

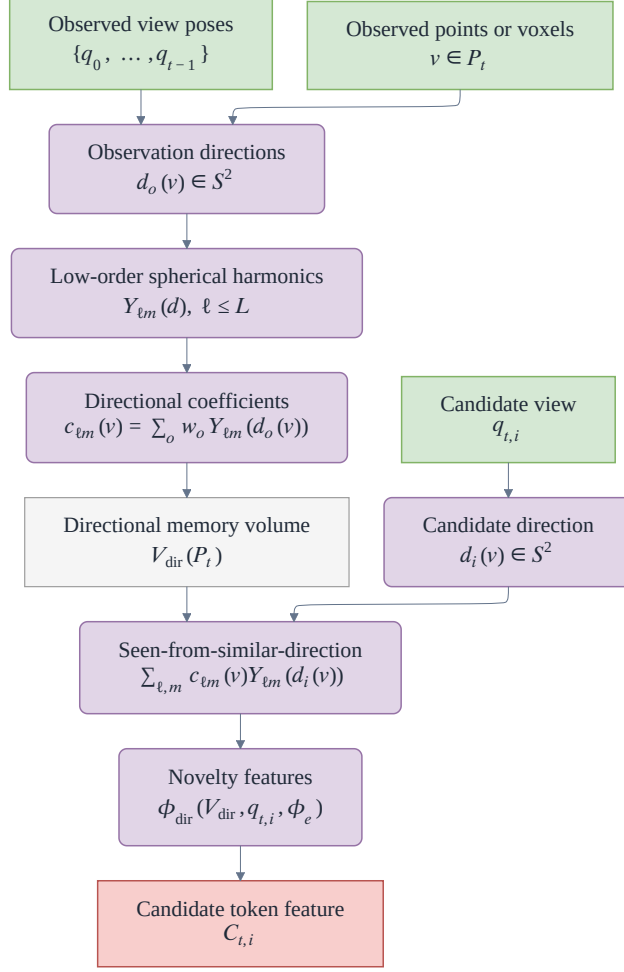


Figure 4: Actor-visible directional memory for target-local view novelty. The figure shows a planned descriptor branch, not an implemented performance result: selected view directions over observed points or voxels are summarized as low-order directional coefficients or second moments, and each valid candidate reads the memory to produce a candidate token feature for Q_H .

The row-level descriptor assembled for candidate scoring is

$$\mathbf{x}_{t,i} = \text{concat}(\mathbf{h}_{t,i}^{\text{pose+rel}}, \mathbf{h}_{t,i}^{\text{geom}}, \mathbf{h}_{t,i}^{\text{valid}}, \mathbf{h}_{t,i}^{\text{prov}}, \mathbf{H}_t)$$

with relative pose/relation features, candidate-geometry token $\mathbf{h}_{t,i}^{\text{geom}}$, hard-mask/reason embedding $\mathbf{h}_{t,i}^{\text{valid}}$, provenance embedding $\mathbf{h}_{t,i}^{\text{prov}}$, and selected-history summary $\mathbf{H}_t$. The selected target enters through $\mathbf{h}_e^{\text{tgt}}$ and candidate-target query

pools. This symbolic view keeps the table below as a provenance checklist rather than the model definition.

Descriptor block	Fields	Failure prevented
Target descriptor and token	OBB geometry, class/confidence, selector rank, projection, semidense/EVL support, EVL coverage, source mode.	Keeps V1 target input separate from GT-EVAL matching and GT OBB crops.
Candidate self pose	Reference-pose-relative transform, continuous 6D relative rotation, height/up/frustum scalars, optional LFF controls.	Keeps each finite action row interpretable and invalidity hard-masked.
Candidate-target relation	Candidate-local target displacement, range, bearing, elevation, target-frustum overlap, support change, optional EVL/crop read.	Forces conditioning on the selected target instead of scene-coverage shortcuts.
Candidate-candidate and history relations	Query-local relative poses, angular separation, duplicate indicators, selected-view memory, valid-count features.	Supports row-equivariant interaction and duplicate-row diagnostics.
Feature-bank joins	Point/voxel/ray-cell id, feature-model id, compression version, observation count, lineage, uncertainty, extent status.	Prevents mixing logged RGB/DINO with counterfactual selected-depth geometry.
Relation encodings	Query-local target/candidate/history/support relations, source tags, optional attention-bias embeddings.	Transfers QCNet-style geometry without motion-forecasting objectives.
Directional memory	Target-local $\mathbb{S}^2$ moments, optional SH coefficients, directional novelty, selected-view support weights.	Keeps accumulated visibility separate from pose and sampler features.
Provenance	Sampler family, source role, descriptor version, feature-source id, compression id, and missing-modality masks.	Lets ablations test shortcut risk instead of hiding provenance inside pose geometry.

Table 8: Actor-visible descriptor blocks and the leakage boundary before heavier scene encoders are evaluated.

Validation TODO: Before promoting any feature-bank, point/sparse, radiance-field, or recurrent scene encoder to thesis-core status, run EVL-extent, source-dropout, density, row-shuffle, mask-isolation, storage, and runtime ablations against the simpler descriptor controls.

Source: autoresearch thesis-lit-review Iterations 16, 24, 25, and 26; local EFM3D and Deja View literature refresh

Gate: representation ablation evidence

Validation TODO: Resolve the current wording/code tension around outside-EVL extent: some rollout contracts list outside-EVL support as a hard reason, while current VIN evidence treats low EVL coverage as a diagnostic/support feature. The thesis-core rule should be: infeasible pose, missing oracle/evaluation sample, or empty target crop is hard invalidity; low local EVL support alone is a model feature unless it blocks evaluation.

Source: docs/contents/thesis/questions.qmd:21; aria_nbv/aria_nbv/vin/experimental/model.py:848; aria_nbv/aria_nbv/vin/model_v3.py:1465

Gate: rollout invalidity audit before first Q_H training run

4.3 Candidate and Replay Contract

Each decision state carries a finite candidate table $\mathcal{Q}_t$, hard mask $\mathbf{m}_t$, invalid-reason vector $\boldsymbol{\rho}_t$, and the target descriptor $\boldsymbol{\varphi}_e$. It also stores selected-view history and remaining budget. The admissible action is a valid candidate row index:

$$\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_q}, \quad \mathcal{A}_t = \{i \in \{1, \dots, N_q\} : m_{t,i} = 1\}$$

Selecting a candidate means choosing a valid index $a_t = i \in \mathcal{A}_t$ for the transition. Oracle rendering follows the calibrated depth-rendering contract, so camera-frame and rasterizer conventions are part of the label contract rather than model input [Met25c]. All valid candidates may be rendered at the oracle layer to score one-step labels, while the rollout writer separately persists selected/parent depth at a canonical configured resolution as actor-history state for successor Q_H encoders.

After selection, acquired geometry is added to the current geometry:

$$\mathcal{P}_{t+1} = \mathcal{P}_t \cup \mathcal{P}_{q_t}$$

The representation-level transition also updates the sparse ray-aware memory:

$$\mathbf{M}_{t+1}^{\text{ray}} = \text{Fuse}(\mathbf{M}_t^{\text{ray}}, \mathcal{P}_{t,i}^{\text{cand}}, \mathcal{R}_{t,i}^{\text{sel}})$$

This update can add selected surface evidence, ray-carve selected free space, convert unknown cells into observed-free or observed-surface cells, update support counts and uncertainty, and refresh target-local directional memory. It cannot attach a visual descriptor to newly selected counterfactual geometry unless a corresponding actor-visible RGB observation exists; those cells instead carry a selected-depth geometry source and a missing-visual-descriptor mask.

The next candidate table $\mathcal{Q}_{t+1}$ is regenerated from updated geometry, selected-view history, and remaining horizon metadata with the same logged mixture families, while root local EVL evidence remains fixed unless a later ablation explicitly recomputes it. The current target-conditioned mixture vocabulary contains forward/local candidates, target-bearing candidates, lateral target-bypass candidates, bounded orientation jitter, and per-row strategy provenance. The older radial free-shell sampler from the seminar paper is retained as a historical upper-bound or stress ablation, not as the default target-conditioned candidate distribution.

Candidate provenance is a model input only through typed scalar or embedding channels. The row stores `strategy_id`, `position_id`, `mixture_id`, `sampler_probability`, target-distance/bearing diagnostics, motion-realism diagnostics, and invalid-reason bits. The training reader may embed these as candidate-family tokens, but the model must still pass row-shuffle and duplicate-row tests: the family label explains how the row was sampled, not an ordering prior.

Candidate order has no semantics, so shuffled-candidate evaluation is required. The descriptor section defines the candidate self token, query-local relation encodings, and target-local directional memory used by the model. The replay contract stores the canonical facts those descriptors are derived from: poses in documented frames, selected-view lineage, candidate-family provenance, target/support counters, masks, and reason codes. This keeps row descriptors reproducible without making their current tensor encoding the immutable data format.

The minimum replay row contains scene/snippet/target/step identifiers, counterfactual state, target descriptor, candidate table, masks, invalid reasons, selected action, target reward, successor state, successor candidates, successor masks, and policy/seed/sampler metadata. This row reproduces the mask, selected transition, value target, and oracle re-evaluation.

4.4 Architecture Contract and Geometric Acceptance Tests

The geometric-learning rationale in Section 2.2 becomes replay-field acceptance tests. Each module must answer a symmetry or provenance requirement: candidate rows need row-level permutation equivariance; invalid and padded rows need mask isolation; candidate, target, and history geometry need local frames; and actor/oracle fields need separate provenance. The task is gravity aligned and egocentric, so exact global $SO(3)$ or $SE(3)$ equivariance is an ablation claim, not the default system claim.

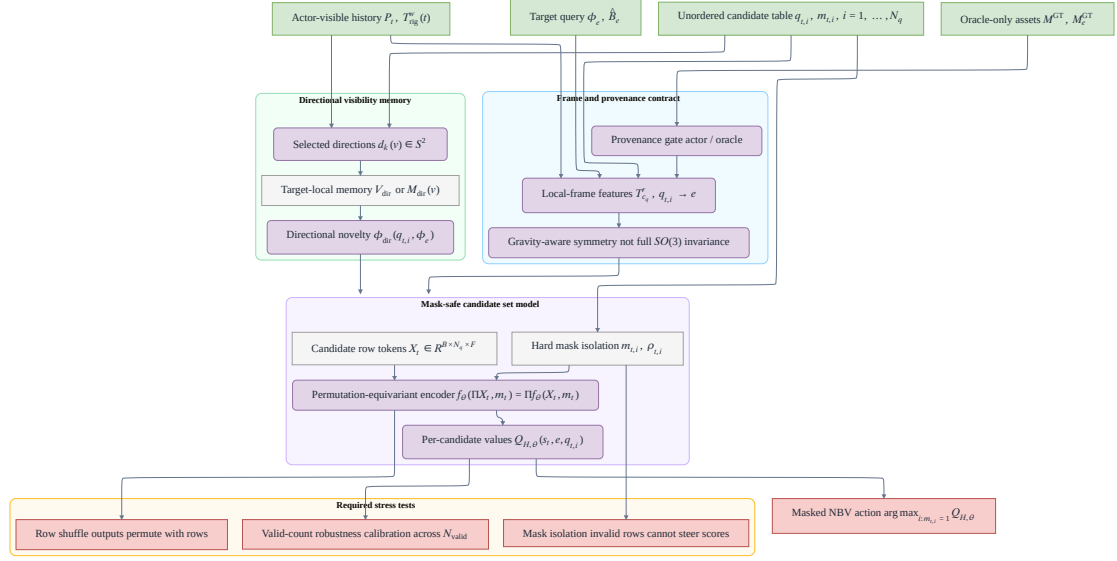


Figure 5: Minimum symmetry and provenance contract for the finite-candidate Q_H model. The contract requires row-level equivariance, mask isolation, local-frame geometry, target-local directional memory, and oracle/actor provenance gates; it does not claim exact global SE(3) equivariance.

The first model family should therefore use scalar invariant and local-frame relative features before heavier equivariant tensor machinery. Relative transforms such as $\xi_{r_t,i}^{\text{rel}}$ and candidate-local target vectors such as $\mathbf{R}_{t,i}^\top(\mathbf{c}_e - \mathbf{c}_{t,i})$ are deliberate gauge choices: they remove irrelevant global-coordinate dependence while preserving yaw, elevation, distance, gravity/up, frustum, and approach-direction signals needed for visibility. Directional history is a separate object. A selected view direction belongs on $\mathbb{S}^2$ and should be stored as a target-local histogram, second-moment matrix, or low-order spherical-harmonic memory rather than being merged into generic pose features. This separation follows the geometric-learning distinction between physical symmetries and useful task structure, and it protects the interpretation of each ablation: a QCNet-style relative positional bias tests candidate-candidate geometry, while directional memory tests whether the target has already been observed from similar directions [Bro+21, e3n25, Zho+23].

The architecture acceptance tests are as important as validation loss. Row-shuffle tests must satisfy $f_\theta(\Pi X_t, m_t) = \Pi f_\theta(X_t, m_t)$ up to numerical tolerance for every per-candidate output used by selection. Mask tests must show that invalid rows cannot alter valid scores except through explicit valid-count or support features. Valid-count and duplicate-row stress tests check whether attention normalization has corrupted absolute target-specific RRI calibration. Only after the independent scorer and candidate-to-state query controls pass these tests should DeepSets context, masked Set Transformer interaction, Fisher/SCONE overlap bias, QCNet-style local relative positional encoding, or EGNN-style candidate graphs be credited as architectural gains [GML22, JLD24, Lee+19, SHW21, Zah+17].

The token design follows the same separation of concerns. State tokens summarize known actor-visible evidence before the next view; candidate tokens describe one finite action and its mask/reason code; relation tokens describe target/candidate/history geometry; label tokens store GT target crops, target gains, endpoint metrics, and TD targets outside the actor graph. The first model should therefore start with a per-candidate scorer and candidate-to-state cross-attention;

pooled set context or masked candidate-candidate attention is an ablation only when interaction is needed.

Research TODO: Treat Fisher/SCONE overlap attention and QCNet-style relative encodings as ablation hypotheses until row-shuffle, mask-isolation, and paired oracle policy evidence show they improve target-specific endpoint gain over simpler controls.

Source: autoresearch thesis-lit-review report; docs/contents/theory/candidate_view_dependence.qmd

Gate: A2/A3/A4 architecture ablation tables

4.5 Finite-Candidate Value Model

The value-model hypothesis is that a masked finite-candidate model can recover positive oracle-lookahead headroom from actor-visible state. Q_H maps each valid candidate row to a finite-horizon value using actor-visible scene, target, selected-history, budget, candidate, mask, and reason-code features. Its outputs select actions, but thesis evidence comes from oracle re-scoring of the selected trajectories.

The model class follows the structure of the decision problem. The action space is a masked finite set of candidate views, each defined relative to a target, selected history, and partially observed geometry. Geometric deep learning supplies vocabulary for these regularities without committing the thesis to a full equivariant tensor network [Bro+21]. Candidate-row permutation requires equivariant per-candidate outputs; local camera and target frames reduce dependence on global coordinates; $\mathbb{S}^2$ visibility memory records where the target has already been observed; and the target record acts as the query that determines which reconstruction errors matter.

The architecture critique is therefore simple: the first serious model should not be a monolithic transformer over every available tensor or over arbitrary candidate rows. The clean object is a calibrated one-step target utility field plus an uncentred finite-horizon residual from candidate-to-state evidence. The one-step field owns immediate target-gain calibration; the residual owns downstream effects from candidate regeneration, selected-history geometry, occlusion, free/unknown evidence, and support overlap. Unrelated sampled rows should not change the

physical value of candidate q_i except through explicitly modeled support or policy-context ablations.

$$b_{\psi,i} = f_{\psi}^{\text{1-step}}(s_t^{\text{cf0}}, \boldsymbol{\varphi}_e, q_{t,i}), \quad \delta_{\theta,i}^H = g_{\theta}(\mathbf{x}_{t,i}, \{\mathbf{h}_e^{\text{tgt}}, \mathbf{M}_t^{\text{ray}}, \mathbf{H}_t, \mathbf{E}_0^{\text{EVL-local}}\}, h_t), \quad Q_{H,\theta,i} = b_{\psi,i} + \delta_{\theta,i}^H$$

The residual can be kept small with a norm penalty or dataset-level regularization, but it should not be exactly mean-centred within each sampled candidate table. Per-set centering changes absolute Q values when duplicate or unrelated valid rows are added, which is unsafe for TD targets. Candidate-candidate interaction remains useful for policy logits, diversity, or top- k selection ablations, but the default finite-horizon value head stays in continuous target-gain units. Recent object-centric view-planning work reinforces this separation: target-centric visibility and feasibility should be explicit scoring factors, while difficulty, reachability, budget, and object saturation must be reported separately because they can change planner rankings and failure modes [Hu+26, Pan+26].

Design principle		Adopt in Q_H	Reject or defer
Calibrated	absolute field	Keep a one-step target-gain head that can be evaluated candidate-by-candidate under fixed masks.	Letting attention over unrelated rows refine the absolute immediate-RRI label.
Finite-horizon	residual context	Use candidate-to-state cross-attention for the default residual; regularize residual magnitude without exact per-set centering.	A pooled scene embedding with no per-row path, exact candidate-table mean centering for TD values, or unmasked invalid-row attention.
Typed relative	geometry	Encode target-current-candidate-history relations in local frames; use QCNet-style RPE as an ablation.	Claiming full global SE(3) equivariance for an egocentric, gravity-aligned, frustum-limited task.
Directional	visibility	Represent selected history on S^2 and keep it distinct from generic pose tokens.	Collapsing target-local observability into a scalar distance or sampler family prior.
Support and	difficulty reporting	Report EVL extent, semidense support, reachability, validity, budget, and target saturation bins.	Using support, Fisher/coverage proxies, or target area as the thesis reward instead of oracle target-RRI.
Late geometric	transformers	Treat EGNN, SE(3)-Transformer, and GATr-style equivariant modules as support-encoder or candidate-graph ablations.	Making exact equivariance the first thesis claim before the replay/mask/target-RRI contract is stable [Bre+23, Fuc+20, SHW21].

Table 9: Architecture critique distilled into a conservative design order. The clean first model preserves a calibrated candidate-local field and allows actor-visible state, history, and geometry context to explain finite-horizon residual value under explicit masks and evaluation gates.

The one-step target scorer is adapted to counterfactual rollout rows rather than reusing the seminar VIN checkpoint unchanged. It remains myopic and predicts immediate target-specific RRI evidence for each candidate; the seminar VINv3 scorer is therefore a control architecture and implementation substrate, not already a target-conditioned finite-horizon Q_H result. Q_H is residual around this calibrated base, with an uncentred continuous residual head as the canonical finite-horizon value definition. The myopic scorer uses the CORAL ordinal-regression interface of Cao et al., adapted to ARIA-NBV’s skewed oracle RRI labels [CMR19]:

$$p_{t,i,k}^{\text{CORAL}} = \sigma(o_{t,i,k}^{\text{CORAL}}), \quad k = 0, \dots, K-2$$

$$\pi_{t,i,k}^{\text{CORAL}} = p_{t,i,k-1}^{\text{CORAL}} - p_{t,i,k}^{\text{CORAL}}, \quad p_{t,i,-1}^{\text{CORAL}} = 1, \quad p_{t,i,K-1}^{\text{CORAL}} = 0$$

$$\hat{r}_{\psi}^e(s_t^{\text{cf0}}, \varphi_e, q_{t,i}) = \sum_{k=0}^{K-1} \pi_{t,i,k}^{\text{CORAL}} u_k$$

ARIA-NBV’s adaptation is in the binning and decoding around that interface. Continuous oracle target gains are fitted to empirical quantile edges $\tau_1 \leq \dots \leq \tau_{K-1}$, and each sample receives the ordinal label $y = \sum_{j=1}^{K-1} \mathbb{1}[r^e > \tau_j]$. CORAL levels are threshold indicators $\ell_k = \mathbb{1}[y > k]$ for $k = 0, \dots, K-2$. The scorer then decodes logits both as cumulative probabilities $P(y > k) = \sigma(o_k)$ and as a ranking proxy $E[y] = \sum_k \sigma(o_k)$; when calibrated bin representatives are available, the expectation over u_k maps the ordinal distribution back to target-gain units. This preserves the VIN-NBV ordinal-ranking precedent while making calibration, bin drift, and residual Q_H recovery explicit ARIA-NBV diagnostics.

Training is staged to preserve the residual interpretation: train and calibrate $\hat{r}_{\psi}^e$, then freeze or slow-finetune it while fitting residual Q_H in continuous return units with Huber or distributional/quantile losses, and finally ablate whether end-to-end fine-tuning improves oracle-evaluated policy performance. CORAL remains the one-step ranking/calibration interface; finite-horizon Q_H must preserve the metric structure of additive returns. This also keeps the educational story clean: VINv3 answers “which single candidate looks good now?”, while Q_H answers “which first action has the best bounded downstream target gain under the same finite candidate and mask contract?”.

The descriptor protocol in Table 8 supplies the model input: actor-visible target token, scene/support memory, selected-history state, candidate self tokens, query-local relations, ray queries, hard masks, and reason codes. The default candidate-query encoder is candidate-to-state cross-attention, so each physical candidate reads the same fixed target, map, history, budget, and local-EVL tokens without unrelated candidate rows redefining its absolute value:

$$\mathbf{u}_{t,i} = \text{CrossAttn}_{\theta}(\mathbf{x}_{t,i}, \{\mathbf{h}_e^{\text{tgt}}, \mathbf{M}_t^{\text{ray}}, \mathbf{H}_t, \mathbf{b}_t, \mathbf{E}_0^{\text{EVL-local}}\})$$

The candidate-query architecture in Figure 6 expands the same contract into the architecture used for the thesis hypothesis and ablation ladder.

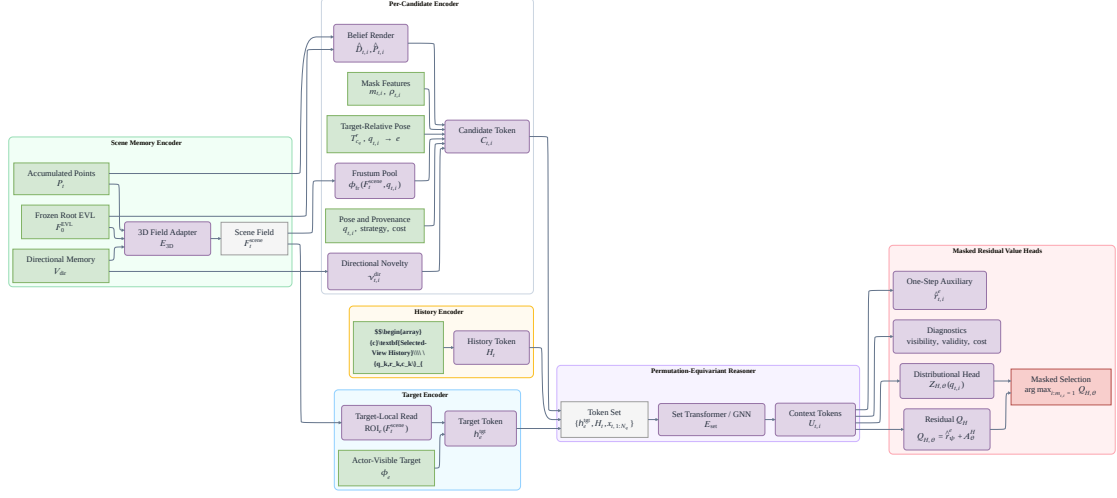


Figure 6: Candidate-query Q_H architecture sketch. Actor-visible EVL evidence, accumulated geometry, ray-aware memory, target descriptors, selected-history summaries, candidate geometry, masks, and directional memory feed candidate-to-state value queries; residual finite-horizon values, one-step auxiliary scores, and diagnostics are then decoded only over valid candidate rows. Oracle labels supervise training and evaluation outside this actor-input graph.

No-interaction candidate MLP scoring and candidate-to-state cross-attention are required baselines before attributing gains to masked Set Transformer interaction or QCNet-style RPE. For immediate target-specific RRI, the physical oracle label of candidate q_i does not change when unrelated rows are added to $\mathcal{Q}_t$. Candidate interaction can therefore corrupt absolute calibration if it replaces the independent scorer. The safer finite-horizon ablation lets candidate context influence policy logits, diversity, or a separately regularized residual, but not an exact per-set mean-centred TD value.

The value head is an uncentred residual decomposition over valid actions, with residual regularization rather than exact candidate-table centering:

$$Q_{H,\theta}(s_t^{\text{cf0}}, \varphi_e, q_{t,i}) = \hat{r}_\psi^e(s_t^{\text{cf0}}, \varphi_e, q_{t,i}) + \delta_{\theta,i}^H(s_t^{\text{cf0}}, \varphi_e, \mathbf{H}_t), \quad \mathcal{L}_\delta = \lambda_\delta \frac{1}{|\mathcal{A}_t|} \sum_{j \in \mathcal{A}_t} (\delta_{\theta,j}^H)^2$$

Model role	Content
Hypothesis model	adapted target-conditioned myopic scorer, candidate-to-state residual Q_H in continuous return units, hard masks/reasons, matched-budget oracle re-scoring
Required controls	independent candidate MLP, calibrated myopic scorer, and candidate-to-state cross-attention without candidate-candidate self-attention
Ablations	pooled DeepSets context, masked Set Transformer policy context, QCNet-style candidate-local RPE, Fisher/SCONE support-overlap attention bias, $\mathbb{S}^2$ memory variants, EGNN-style candidate graph, privileged-teacher distillation, distributional Q_H heads
Architecture ladder	A0 independent scorer; A1 candidate-to-state cross-attention; A2 pooled DeepSets context; A3 masked Set Transformer policy/context ablation; A4 query-local relative bias; A5 Fisher/SCONE overlap bias; A6 distributional or quantile Q_H
Bridges	Hestia-style target-then-pose policies, online discrete interaction, external mesh/oracle-compatible simulators, sparse/point backbones

Table 10: Value-model hypothesis, controls, and ablations. Dense GT candidate renders may supervise later ablations, while learned policy inputs use the configured state and target-task descriptors.

Implementation TODO: Confirm which architecture ladder levels are implemented, planned, or deferred once the final code state is frozen.

Source: advisor handout; proposal method

Gate: method implementation audit

5 Experimental Design

The experiment design evaluates whether the oracle/data-generation pipeline from Chapter 3 and the learned method from Chapter 4 produce leakage-safe target-aware view choices under matched acquisition budgets. It treats data support, target identity, validity masks, replay integrity, and oracle-lookahead headroom as preconditions for interpreting value-model performance.

5.1 Objectives and Hypotheses

The first aim defines target-specific oracle RRI while keeping target selection and model input actor-visible. V1 uses observed or predicted target descriptors φ_e . GT crops, boxes, meshes, and all-candidate renders are restricted to labels, upper bounds, and evaluation. The required evidence is target eligibility, match score, unmatched/ambiguous counts, endpoint $J_e^{(H)}$, separate scene RRI, and acquisition cost.

The render-path boundary in Figure 7 makes this restriction explicit for render-derived training signals.

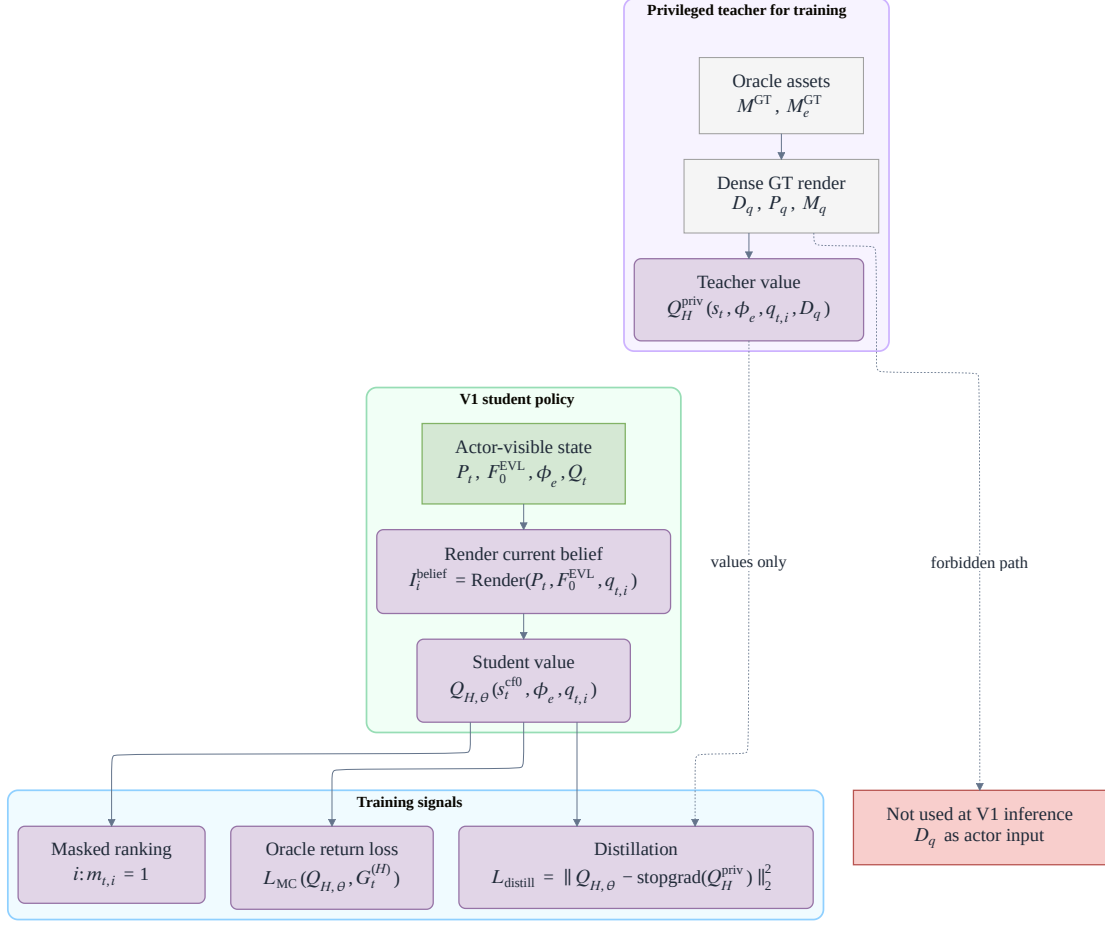


Figure 7: Teacher/student render path for leakage-safe training. The student branch consumes actor-visible state and current-belief render products, while privileged GT meshes, target crops, and dense candidate renders may produce oracle returns, teacher values, or distillation targets only. Dense GT candidate depth is therefore label or teacher evidence, not a V1 actor input.

The second aim trains a VIN-style myopic scorer over the same candidate table that later feeds Q_H . The scorer predicts target RRI from actor-visible scene, target, and candidate features. It is the required learned one-step control, evaluated by rank correlation, top- k oracle hit rate, calibration, selected-candidate oracle RRI, target visibility, invalid fraction, and grouped failures.

The third aim first estimates whether bounded oracle lookahead has headroom over one-step oracle greedy:

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}})$$

Only if this headroom is positive is Q_H expected to recover part of it from offline rollout traces:

$$\eta_Q = \frac{J_e^{(H)}(\pi_Q) - J_e^{(H)}(\pi_{\text{learned-1}})}{J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{learned-1}}) + \varepsilon}$$

Success is measured by oracle-rescored selected actions, not predicted values. If oracle lookahead itself has little headroom, the thesis reports that the current objective and candidate distribution are effectively myopic. If lookahead has headroom but Q_H fails to recover it, the analysis reports whether the limiting factor is target observability, candidate support, rollout coverage, reward definition, or model capacity.

Claim	Primary evidence	Decision rule
Target utility	$J_e^{(H)}$, $G_t^{(H)}$, scene RRI, cost	$J_e^{(H)}$ decides endpoint quality; $G_t^{(H)}$ trains and ranks rollouts.
Input safety	actor-visible φ_e , match score, support, leakage checks	GT is label/evaluation only for the V1 result.
Myopic control	target-rank metrics, selected-candidate oracle RRI, calibration	One-step target scoring is the comparator for Q_H , not the final policy claim.
Planning headroom	Δ_{look} and recovered fraction η_Q	Q_H is meaningful only relative to measured oracle-lookahead headroom.

Table 11: Objective-to-evidence matrix.

5.2 Learning Objective and Replay Evidence

The learned value model is interpreted only after the myopic scorer passes ranking, calibration, and oracle-selected rollout checks, and after the replay store passes support, mask, seed, and successor-table checks. These gates make the evaluation contract a scientific guardrail rather than a post-hoc reporting checklist. Seminar-paper diagnostics such as scene-level RRI ranking, CORAL bin behavior, and offline-cache storage are reported as historical substrate evidence unless they are regenerated under the target-task sampler and rollout-store protocol.

The replay pipeline in Figure 8 is the operational boundary for this interpretation: all-candidate labels can train and calibrate the one-step scorer, while Q_H evidence comes only from selected-transition rows whose successor state and successor candidate mask are reproducible.

Data product		Purpose	Minimum evidence
one-step	all-candidate labels	train and calibrate $\hat{r}_\psi^e$	rank correlation, top- k oracle hit, calibration
paired	greedy/lookahead roots	measure Δ_{look}	same root, target, seed, N_q , horizon, and validity masks
stochastic	support traces	avoid fitting Q_H only on greedy states	policy entropy, unique selected chains, target visibility
invalid	near-misses	stress hard masks and failure modes	invalid-reason distribution and with/without-mask diagnostics
target	and candidate strata	test support across target difficulty and candidate families	class/support/area/occlusion bins, per-strategy RRI histograms, successor-table availability

Table 12: Rollout-support coverage. Target, candidate, validity, and successor-table gaps are reported before interpreting value-model differences.

The one-step scorer is trained on all valid candidate rows with oracle immediate target-specific RRI labels. The bootstrapped Q_H backup is trained only on expanded transition rows for which the selected action, successor counterfactual state, successor candidate table, successor mask, and terminal flag are material-

ized. If all-candidate successor expansion is unavailable, non-expanded candidates contribute myopic supervision but not bootstrapped finite-horizon targets.

Evidence surface	Seminar evidence role	Current thesis gate
Oracle RRI labeler	Implemented scene-level depth-render, fusion, and point-mesh scoring substrate.	Target-cropped labels, empty-crop in-depth-render, fusion, and identity-match audit validity, and identity-match audit pass on current manifests.
Candidate sampling	Legacy free-shell candidate generator and pruning diagnostics.	Target-conditioned mixture reports strategy counts, invalid reasons, headroom strata, and hard-turn diagnostics.
CORAL scorer	One-step ordinal RRI calibration and expected-RRI interface.	Target scorer passes ranking/calibration before residual Q_H is interpreted.
Offline stores	Immutable one-step VIN payload proves feasibility of materialized labels.	Selected-transition rollout store reports successor tables, masks, seeds, storage, and replay integrity.
Architecture	VINv3/EVL myopic scorer is the implemented control.	A0-A5 ablation ladder is compared by paired oracle-rescored endpoint gain.

Table 13: Substrate-to-thesis evidence matrix used to avoid treating historical seminar results as final target- Q_H claims.

The finite-candidate value model decodes actions only over valid candidate tokens:

$$\mathbf{u}_{t,i} = (\text{Transformer}_{\theta}(\mathbf{X}_t^{\text{cand}}))_i$$

$$a_t^{\theta} = \underset{i:m_{t,i}=1}{\operatorname{argmax}} Q_{H,\theta}(s_t^{\text{cf0}}, \boldsymbol{\varphi}_e, q_{t,i})$$

The initial backup is fitted masked Double-Q [HGS15]. Here $d_t = 1$ at horizon termination, budget termination, or when no valid successor action exists:

$$i^{\star} = \underset{i:m_{t+1,i}=1}{\operatorname{argmax}} Q_{H,\theta}(s_{t+1}^{\text{cf0}}, \boldsymbol{\varphi}_e, q_{t+1,i})$$

$$y_t = r_t^e + \gamma(1 - d_t)Q_{H,\theta}(s_{t+1}^{\text{cf0}}, \boldsymbol{\varphi}_e, q_{t+1,i^{\star}})$$

The replay dataset $\mathcal{D}$ contains selected-action transition rows with state, action, immediate target reward, successor state, validity masks, and terminal flags:

$$\mathcal{L}_Q(\theta) = \frac{1}{|\mathcal{D}|} \sum_{(s,a,r,s') \in \mathcal{D}} m_{t,a} (Q_{H,\theta}(s_t^{\text{cf}0}, \varphi_e, q_{t,a}) - y_t)^2$$

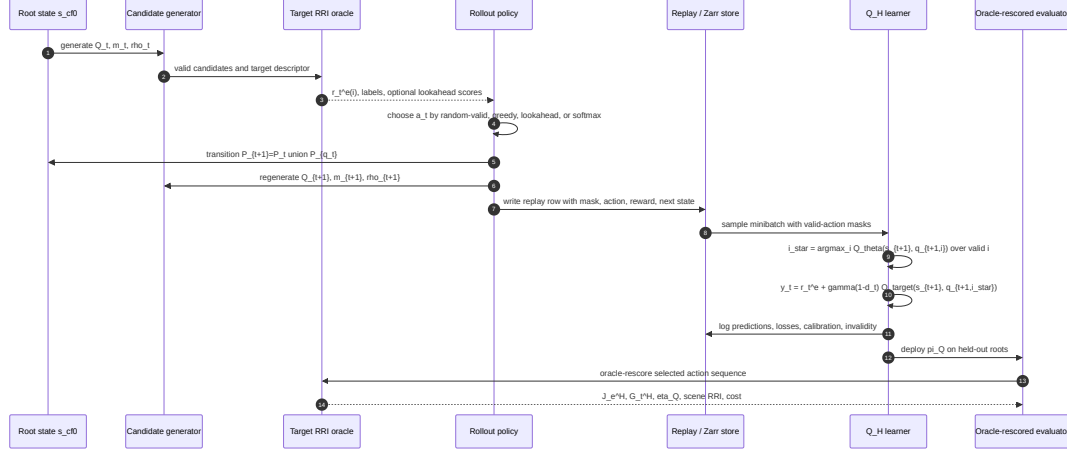


Figure 8: Selected-transition replay contract for Q_H . All valid candidates can receive one-step oracle target labels, but bootstrapped finite-horizon targets require materialized selected actions, successor counterfactual state, regenerated successor candidate tables, masks, terminal flags, and masked Double-Q targets.

5.3 Policy Comparison

All selected actions are oracle-evaluated under the same acquisition and candidate budgets. Equal budget means equal selected-view horizon H , candidate count N_q , candidate-generation distribution, and validity constraints; path length, runtime, and oracle evaluation count are reported separately. Coverage is reported against the planned full scale bar of 100 GT mesh ASE scenes and 4,608 snippet windows from the current thesis contract, or against an explicit scene-level held-out subset if scale is blocked [Met25a]. Final splits are scene-level; sample-level splitting across snippets from the same scene is not valid for final claims.

Validation TODO: Replace planned scale bars with generated manifest statistics for the current one-step store and rollout store; historical seminar subset counts are not current thesis evidence unless regenerated. Source: docs/typst/seminar_paper/sections/04-dataset.typ; docs/typst/seminar_paper/sections/09a-evaluation.typ; current dataset manifests
Gate: final experiment tables

Policy	Actor	in-	GT	deci-	H	Role
	put			sion		
π_{rand}	yes		no		1	lower reference over valid candidates
$\pi_{\text{learned-1}}$	yes		no		1	myopic learned target scorer
$\pi_{\text{oracle-1}}$	no		yes		1	one-step oracle upper bound
$\pi_{\text{oracle-look}}$	no		yes		H	cumulative-RRI headroom estimate
π_Q	yes		no		H	learned recovery over myopic scoring when headroom is positive

Table 14: Leakage-aware policy comparison. Report $J_e^{(H)}$, $G_t^{(H)}$, scene RRI, cost, invalidity, runtime, and coverage for each row.

Policy comparisons are paired by root snippet, target, candidate seed, candidate budget, and horizon. Report mean, median, bootstrap confidence intervals, and per-scene win rates for $J_e^{(H)}$, $G_t^{(H)}$, and invalidity; scene-level failures are reported separately from global averages.

5.4 Failure Interpretation

The failure interpretation is part of the research contract. If geometry or oracle labels fail, the contribution becomes a validation and one-step-scoring study. If target matching is sparse or ambiguous, target-specific RRI is reported only on validated subsets with unmatched counts and acceptance filters. If Δ_{look} is near zero, the thesis reports no measurable non-myopic headroom for the evaluated split, target set, horizon, branch factor, and candidate distribution. The next diagnosis is target matching, candidate support, and supervision scale; added model complexity is justified only after those evidence gaps are ruled out. Scaling and online-discrete tests remain interpretable only if they preserve target-specific RRI supervision.

Validation TODO: Replace planned evidence gates with actual result tables, figures, and failure cases after M1-M5 runs are complete.

Source: proposal objectives/evaluation; advisor handout

Gate: final experiments

6 Results

This chapter is reserved for generated evidence from the final rollout manifests, scorer runs, value-model ablations, and paired oracle re-evaluation. Until those artifacts are frozen, planned scale bars, historical seminar subsets, and exploratory diagnostics remain in the experimental-design chapter or appendices as evidence gates rather than final results.

Validation TODO: Replace this scaffold with manifest-backed result tables: target eligibility and match coverage, candidate validity and invalid-reason distributions, one-step scorer ranking/calibration, measured oracle-lookahead headroom, Q_H recovery, endpoint target gain, runtime, storage, and failure strata.

Source: thesis migration plan

Gate: final experiment manifest and result-table generation

7 Discussion

The discussion will interpret failures and gains only after the prerequisite support checks are available. If oracle lookahead has little headroom, the correct conclusion is about the evaluated target set, candidate distribution, horizon, and support regime, not about neural planning capacity in general. If headroom exists but Q_H fails to recover it, the diagnosis should separate target identity, candidate validity, scene-representation support, replay coverage, mask handling, calibration, and model capacity.

Representation conclusions should preserve the actor/oracle boundary. EFM3D/EVL can be credited as Aria-native local target and support evidence; sparse ray-aware memory, semidense support, and logged DINO-on-point descriptors should be discussed as actor-visible representation ablations. CubeRCNN-style ROI features remain detector/appearance baselines unless an ARIA/ASE target-evidence contract comparable to EFM3D is demonstrated.

Research TODO: After final experiments, move only evidence-backed limitations and future-work claims here; keep implementation recipes in Quarto implementation notes.

Source: thesis migration plan

Gate: final results pass

8 Conclusion

The expected final contribution is a reproducible target-aware finite-candidate view-selection study, not a broad claim that continuous reinforcement learning has been solved for egocentric reconstruction. The thesis tests a single planning hypothesis: after oracle target-task sampling and target-specific supervision are defined, finite-candidate oracle lookahead should expose whether non-myopic target-specific RRI headroom exists, and a learned finite-horizon value model should recover part of that headroom under matched oracle re-evaluation.

The scope limit is central to the conclusion. The thesis does not claim full continuous-control NBV, online RL, real-device deployment, or replacement of target-specific point-mesh RRI by coverage, uncertainty, or semantic proxy objectives. Online discrete interaction, continuous target-then-pose policies, simulator-backed actor-critic, SceneScript, VLM planning, sparse/point backbones, and 3DGS control are escalation studies; they enter only if they preserve the finite-candidate target-specific RRI comparison.

If Δ_{look} is positive and Q_H recovers part of it under oracle-rescored selected trajectories, the thesis supports the claim that target-conditioned finite-candidate planning contains learnable non-myopic structure in the evaluated ASE/EVL regime. If headroom is near zero, the thesis instead reports a negative planning result for the evaluated candidate generator, horizon, target set, and split. If headroom exists but Q_H does not recover it, the failure analysis separates target observability, candidate support, rollout coverage, reward definition, and model-capacity explanations.

Research TODO: Rewrite the conclusion after final evidence is known. Preserve the conditional success/failure structure rather than forcing a positive planning claim.

Source: proposal risk control; advisor handout

Gate: final writing freeze

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Appendix A: Supplementary Material

Implementation TODO: Add supplementary derivations, camera-convention details, full seminar-substrate adaptation notes, generated manifest tables, and implementation-flow figures.

Source: thesis peer-review pass; seminar paper oracle/CORAL/offline-cache sections

Gate: final appendix pass

9 Draft Intake and Open Work

Archived source note: This chapter preserves substantive planning material from the retired proposal and advisor handout. It is part of the thesis seed, not final thesis prose. Each item should either graduate into the main chapters, move to an appendix, or be removed after its source decision is resolved.

Source: archived proposal and advisor handout

9.1 Schedule and Risk Control

The planned thesis window from the proposal ran from 29 April 2026 to 30 September 2026. The roadmap owns the live schedule; this appendix keeps only the exit-condition logic. The work must progress from source-aligned proposal scope, through data/oracle validation, one-step target scoring, target-task rollouts, oracle-lookahead headroom, and Q_H recovery, before bridge designs such as online discrete control or continuous policies are promoted. Final writing freezes only after manifests, coverage, failure cases, reproducible configs, and PDF artifacts are generated.

Conflict: The absolute dates are retained from the proposal source but must be checked against the current roadmap before they appear as final thesis text.

Source: proposal schedule; current roadmap

Gate: roadmap synchronization

9.2 Preliminary Thesis Outline

The active thesis body now uses the expanded chapter graph: foundations, oracle/data generation, method, experimental design, results, discussion, and conclusion. That split keeps oracle state, target-RRI labels, candidate validity, and rollout/replay stores out of the learned-method chapter, while Chapter 04 owns the scene representation, descriptor/query pools, candidate-row architecture, and finite-horizon Q_H model.

Research TODO: Freeze whether any remaining appendix-only draft material should be promoted into the active 01–08 chapter graph.

Source: proposal outline; current main.typ chapter graph

Gate: thesis structure freeze

9.3 Open Decisions and Questions

Open advisor decisions are the final scene-level split; symbolic target-match thresholds ($\tau_\mu, \tau_{\text{gap}}, \tau_{\text{support}}$); the CORAL-to- Q_H interface; the initial actor-visible crop descriptor ablation; the final evidence scale and subset rule; and whether any external or online scaling substrate preserves comparable mesh/oracle target-specific RRI labels. These choices affect the strength and scope of the final evidence, not the target-conditioned finite-candidate question.

Open decision: Lock final scene-level split and acceptable scale fallback.

Source: advisor handout; roadmap

Gate: M4/M5

Open decision: Lock target-match thresholds, ambiguity gap, target eligibility, and target-invalid reporting.

Source: proposal problem; advisor handout

Gate: RQ2 protocol

Open decision: Lock whether the myopic scorer freezes, slow-finetunes, or end-to-end fine-tunes when fitting residual Q_H .

Source: advisor value-model section

Gate: Q_H training plan

Open question: Can a privileged critic use GT mesh, OBB, or segmentation cues without weakening the non-privileged learned-policy result?

Source: advisor handout

Gate: critic/surrogate decision

Open question: When, if ever, do RGB, semantics, or 3DGS enter the thesis core rather than a bridge study?

Source: proposal and advisor follow-up sections

Gate: M6 scope

Implementation TODO: Validate that Rerun examples, replay integrity, shuffled-candidate evaluation, duplicate-row robustness, and valid-count sensitivity are available for final evidence reporting.

Source: advisor evidence contract

Gate: evaluation implementation audit

9.4 Future Bridges Retained from Source Material

Continuous control, external simulators, 3DGS control, SceneScript, VLM planning, IQL, CQL, BCQ, sequence decoding, soft/energy policies, PPO, SAC, privileged-teacher distillation, distributional Q_H heads, EGNN candidate graphs, sparse/point backbones, and Hestia-style target-then-pose policies remain future or post-M5 ablation material unless the finite-candidate rollout store is stable and preserves the target-specific RRI comparison.

Research TODO: Route each bridge to final discussion, appendix, or deletion after the implemented evidence is known. Do not promote these bridges to thesis-core claims without matched target-specific RRI supervision and oracle re-evaluation.

Source: proposal method; advisor adoption ledger

Gate: discussion/future-work pass

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Ehrenwoertliche Erklaerung

Ich versichere, dass ich diese Masterarbeit selbststaendig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

Muenchen, TODO: Abgabedatum

Jan Duchscherer